M

Magnetic Bearings (*see* Bearings)

Measurement* (*see also* Condition Monitoring; Control Systems)

Temperature Measurement

Measurement of temperature is generally considered to be one of the simplest and most accurate measurements performed in engineering. The desired accuracy in the measurement can be obtained, however, only by observing suitable precautions in the selection, installation, and use of temperature-measuring instruments and in the proper interpretation of the results obtained with them.

Four phenomena form the basis for most measuring instruments:

- Change in physical dimensions or characteristics of liquids, metals, or gases
- Changes in electrical resistance
- Thermoelectric effect
- Radiant energy

The following types of instruments are available for use under appropriate conditions:

- Liquid-in-glass thermometer
- Resistance thermometer
- Thermocouple thermometer
- Filled-system thermometer
- Bimetallic thermometer
- Radiation thermometer
- Optical pyrometer

Liquid-in-glass thermometer

A liquid-in-glass thermometer consists of a thin-walled glass bulb attached to a glass capillary stem closed at the opposite end, with the bulb and a portion of the stem filled with an expansive liquid, the remaining part of the stem being filled with the vapor of the liquid or a mixture of this vapor and an inert gas. Etched on the stem is a scale in temperature degrees so arranged that, when calibrated, the reading corresponding to the end of the liquid column indicates the temperature of the bulb. The three types of liquid-in-glass thermometers (Fig. M-1) are partial-immersion, total-immersion, and complete-immersion thermometers.

1. A partial-immersion thermometer is designed to indicate temperature correctly when used with the bulb and a specified part of the liquid column in the stem exposed to the temperature being measured; the remainder of the liquid column

* Source: Demag Delaval, USA.

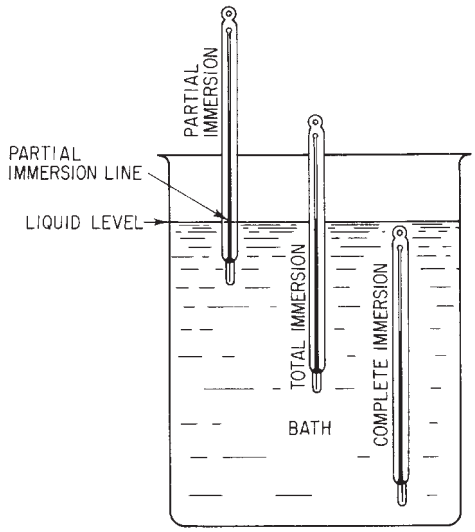


FIG. M-1 Partial-, total-, and complete-immersion thermometers. (Source: Demag Delaval.)

and the gas above the liquid are exposed to a temperature that may or may not be different.

2. A total-immersion thermometer is designed to indicate the temperature correctly when used with the bulb and the entire liquid column in the stem exposed to the temperature being measured and the gas above the liquid exposed to a temperature that may or may not be different.
3. A complete-immersion thermometer is designed to indicate the temperature correctly when used with the bulb, the entire liquid column in the stem, and the gas above the liquid exposed to the temperature being measured.

Tables M-1 and M-2 show National Bureau of Standards (NBS) certification tolerances for laboratory thermometers. The term *tolerance in degrees* means acceptable limits of error of uncertified thermometers. *Accuracy in degrees* is the limit of error to be expected when all necessary precautions are exercised in the use of thermometers. The limits to which NBS certification values are rounded off are shown in the column "Corrections stated to."

The operation of a liquid-in-glass thermometer depends on having the coefficient of expansion of the liquid greater than that of the bulb glass. As a consequence, an increase in temperature of the bulb causes the liquid to be expelled from the bulb, resulting in a rise in position of the end of the liquid column. The capillary stem attached to the bulb serves to magnify this change in volume on a scale.

The most frequently encountered source of error when using liquid-in-glass thermometers is the misuse or complete neglect of the emergent-stem correction. This correction derives from the use of the thermometer with a portion of the stem exposed to a different temperature from that of calibration. A common example is the use of partial immersion of a thermometer calibrated for total immersion. For detailed information on this correction, see the American Society of Mechanical Engineers' *Power Test Codes: Temperature Measurement*.

Resistance thermometer

A resistance thermometer is a temperature-measuring instrument in which electrical resistance is used as a means of temperature measurement. The

TABLE M-1 Tolerances for Fahrenheit Mercurial Total-Immersion Laboratory Thermometers

Temperature Range in Degrees	Graduation Interval in Degrees	Tolerance in Degrees	Accuracy in Degrees	Corrections Stated to
	Thermometers for Low Temperatures			
–35 to 32	1 or 0.5	1	0.1–0.2	0.1
–35 to 32	0.2	0.5	0.05	0.02
	Thermometers Not Graduated above 300°			
32 up to 300	2	1	0.2–0.5	0.2
32 up to 300	1 or 0.5	1	0.1–0.2	0.1
32 up to 212	0.2 or 0.1	0.5	0.02–0.05	0.02
	Thermometers Not Graduated above 600°			
32 up to 212	2 or 1	1	0.2–0.5	0.2
Above 212 up to 600		2	0.5	0.2
	Thermometers Graduated above 600°			
32 up to 600	5	4	0.5–1.0	0.5
Above 600 up to 950		7	1–2	0.5
32 up to 600	2 or 1	3	0.2–1.0	0.2
Above 600 up to 950		6	0.5–1.0	0.2

TABLE M-2 Tolerances for Fahrenheit Mercurial Partial-Immersion Laboratory Thermometers

Temperature Range in Degrees	Graduation Interval in Degrees	Tolerance in Degrees	Accuracy in Degrees	Corrections Stated to
	Thermometers for Low Temperatures			
–35 to 32	1	1	0.3–0.5	0.1
	Thermometers Not Graduated above 300°			
32 up to 300	2 or 1	2	0.2–1.0	0.2
	Thermometers Not Graduated above 600°			
32 up to 212	2 or 1	2	0.2–0.5	0.2
Above 212 up to 600	2 or 1	3	1–2	0.5
	Thermometers Graduated above 600°			
32 up to 600	5 or 2	5.0	1–2	1
Above 600 up to 950		10	2–3	1

instrument consists of a resistor, a resistance-measuring instrument, and electrical conductors connecting the two. The resistor may be metallic (usually in wire form) or a thermistor (a thermally sensitive variable resistor made of ceramiclike semiconducting material).

The basis for resistance thermometry is the fact that most metals and some semiconductors change in resistivity with temperature in a known, reproducible manner. Several materials are commonly employed for resistance thermometers,

the choice depending on the compromises that may be accepted. Although the actual resistance-temperature relation must be determined experimentally, for most metals the following empirical equation holds very closely:

$$R_T = R_0(1 + AT + BT^2) \tag{M-1}$$

where R_T = resistance at temperature T; R_0 = resistance at 0°C; T = temperature, K; and A and B = constants depending on material.

The temperature-resistance function for a thermistor is given by the following relationships:

$$R = R_0 e^k \tag{M-2}$$

$$k = \beta\left(\frac{1}{T} - \frac{1}{T_0}\right) \tag{M-3}$$

where R = resistance at any temperature T, K; R_0 = resistance at reference temperature T_0, K; e = base of napierian logarithms; and β = a constant (which usually has a value between 3400 and 3900, depending on the thermistor formulation or grade).

Types of resistance thermometers

Platinum thermometer. This thermometer is known for its high accuracy, stability, resistance to corrosion, and other characteristics. It has a simple relation between resistivity and temperature, shown in Eq. (M-1).

Precision platinum thermometer. This thermometer is used to define the International Practical Temperature Scale from −297.3 to 1168.3°F. The purity and physical properties of the platinum of which the thermometer is made are prescribed to meet close specifications. Different procedures are used for making precision thermometers to cover different temperature ranges.

Industrial platinum resistance thermometer. The requirements for reproducibility and limit of error for thermometers of this type are lower than those for standard thermometers; so are the manufacturing precautions lowered for these thermometers.

Nickel resistance thermometer. This thermometer has been adapted satisfactorily in industrial applications for a temperature range from −100 to 300°F. The nickel resistance thermometer is less stable than platinum thermometers, but its low cost favors its usage.

Copper resistance thermometer. Copper is an excellent material for resistance thermometers. Its availability in a pure state makes it easy to match with established standards. The resistivity curve of copper is a straight-line function of temperature between −60 and 400°F, and that makes copper resistance thermometers suitable for the measurement of temperature differences with high accuracy. Copper resistance thermometers are reliable and accurate means of temperature measurement at moderate temperature levels.

Thermistors (nonmetallic resistance thermometers). Thermistors are characterized by a negative coefficient of resistivity, and their temperature-resistivity curve is exponential. Modern thermistors are very stable; they have high-temperature sensitivity and very fast response. Because thermistors are high-resistance circuits, the effect of the lead wires is minimized, and regular copper wires can be used throughout the circuit. Noninterchangeability owing to the difficulty of reproducing

TABLE M-3 Typical Characteristics of Resistance Thermometers

	Noble Metal		Nonmetallic
	Precision	Industrial	Thermistor
Sensitivity	0.1Ω/°C	0.22Ω/°F	Varies with units
Precision	±0.001°C	±0.3°F	±0.02°F up to 200°F
Accuracy	±0.01°C	±3.0°F standard ±1.5°F special	±0.5°F standard ±0.2°F special
Response: bare		15 s	Fast
Response: with well		30 s	
Resistance	25.5Ω at 0°C	25Ω at 32°F	Varies with units
Linearity	70.1°C/50°C span	70.1°C/50°C span	Exponential
Range	–452.2 to 1,168.3°F (–269 to 630.74°C)	–297.3 to 1,950°F (–182.96 to 1,064°C)	–100 to 500°F (–75 to 260°C)

	Base Metal		
	10 Ω	Copper, 100 Ω	Nickel, 100 Ω
Sensitivity	0.22Ω/°F	0.22Ω/°F	0.186Ω/°F (0.213Ω/°F)
Precision	±0.1°F	±0.1°F	±0.1°F
Accuracy	±0.5°F standard ±0.2°F special	±0.5°F standard ±0.2°F special	±0.5°F standard ±0.2°F special
Response: bare	20 s	40 s	40 s
Response: with well	60 s	90 s	90 s
Resistance	10 Ω at 77°F	100Ω at 77°F	100 Ω at 77°F
Linearity	Excellent	Excellent	Excellent
Range	–100 to 300°F (–75 to 150°C)	–325 to 300°F (–200 to 150°C)	–100 to 300°F (–75 to 140°C)

resistance properties and the nonlinearity of the resistivity curve limits the use of thermistors.

Information on important characteristics of different classes of resistance thermometers is included in Table M-3.

Accessories. Some forms of Wheatstone-bridge circuits are used for the measurement of temperature with base-metal or industrial platinum resistance thermometers, while the Mueller bridge is used with precision platinum resistance thermometers.

Thermocouple thermometer

A thermocouple thermometer is a temperature-measuring instrument in which the electromotive force developed in a circuit composed of two dissimilar metals is used as a means of temperature measurement. It consists of a device for measuring electromotive force, a sensing element (thermocouple), and electrical conductors operatively connecting the two.

The thermocouple thermometer operates on the principle that an electric current will flow in a closed circuit of two dissimilar metals when the junctions of the metals are at two different temperatures. Thermocouple materials are available for use within the approximate limits of –300 to 3200°F. Platinum is the generally accepted standard material to which the thermoelectric characteristics of other materials are referred. The emf-temperature relations of conventional thermoelements versus platinum are shown in Fig. M-2. Reference tables of

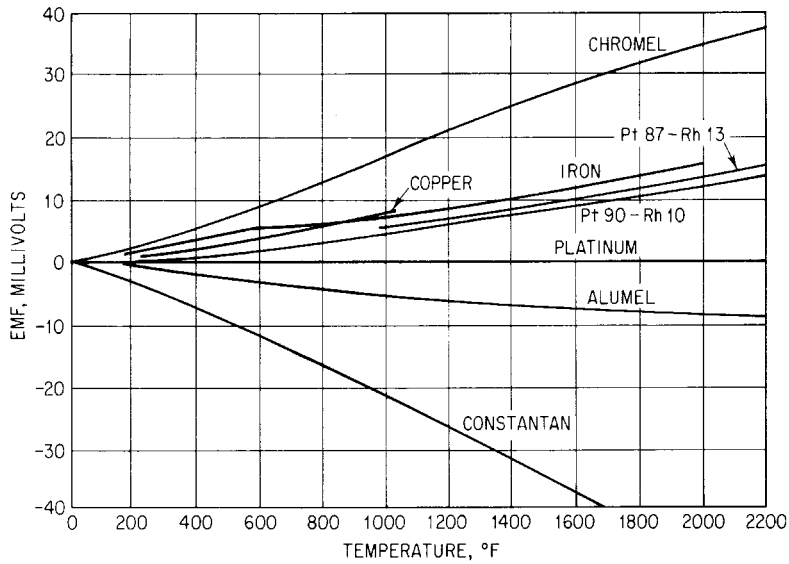


FIG. M-2 emfs of various materials versus platinum. (Source: Demag Delaval.)

TABLE M-4 Limits of Error of Thermocouples

Thermocouple Type	Temperature Range, °C	Limits of Error	
		Standard	Special
T (copper-Constantan)	–184 to –59	. . .	±1%
	–101 to –59	±2%	±1%
	–59 to +93	±0.8°C	±0.4°C
	+93 to +371	±¾%	±⅜%
J (iron-Constantan)	0 to 277	±2.2°C	±1.1°C
	277 to 760	±¾%	±⅜%
E (Chromel-Constantan)	0 to 316	±1.7°C	. . .
	316 to 871	±½%	. . .
K (Chromel-Alumel)	0 to 277	±2.2°C	±1.1°C
	277 to 1260	±¾%	±⅜%
R and S (platinum vs. 13% rhodium-platinum)	0 to 538	±1.4°C	. . .
(platinum vs. 10% rhodium-platinum)	538 to 1482	±¼%	. . .
B (platinum vs. 30% rhodium-platinum vs. 6% rhodium)	871 to 1705	±½%	. . .

temperature versus electromotive force as well as polynomial equations expressing the temperature-voltage relationship for different types of thermocouples are available in technical literature.

The iron-Constantan thermocouple is used most widely in industrial applications. The copper-Constantan thermocouple is used widely in industrial and laboratory thermometry.

The platinum –10 percent rhodium versus platinum (Type S) thermocouple serves as an instrument for defining the International Practical Temperature Scale from 630.74 to 1064.43°C. It is being used in industrial laboratories as a standard for base-metal thermocouples and other temperature-sensing devices.

Table M-4 lists the seven commonly used thermocouples and some of their characteristics.

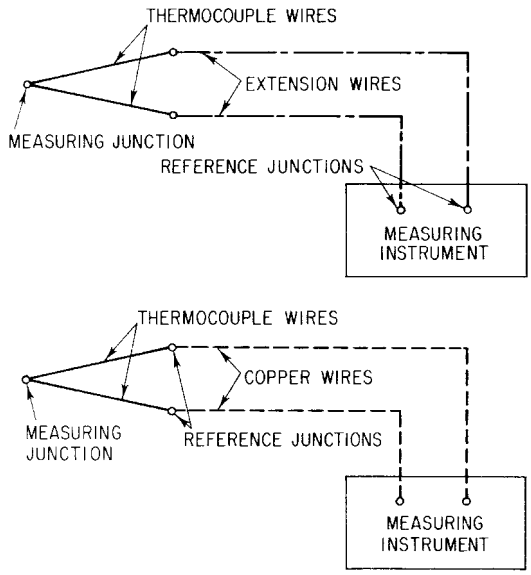


FIG. M-3 Thermocouple thermometer systems. (Source: Demag Delaval.)

The electrical conductors connecting the thermocouple and the measuring instrument may use the actual thermocouple wires, extension wires, or connecting wires (see Fig. M-3). When it is not possible to run the thermocouple wires to the reference junction or to the measuring instrument, extension wires can be used. To assure a high degree of accuracy, extension wires should have the same thermoelectric properties as the thermocouple wires with which they are used. Significant uncertainties are introduced when extension wires are not matched properly. Calibration of the instrument with extension wires helps to minimize these uncertainties. Connecting wires are a pair of conductors that connect the reference junction to the switch or potentiometer. They are usually made of copper. They do not cause uncertainty in measurements when the reference junction is kept at constant temperature, for example, the ice point.

Indicating potentiometers are recommended by the *ASME Power Test Codes* for performance-test work, although recording potentiometers are used for industrial-process temperature measurement.

Thermocouples may be joined in series. The series connection, in which the output is the arithmetic sum of the emfs of the individual thermocouples, may be used to obtain greater measurement sensitivity and accuracy. A series-connected thermocouple assembly is generally referred to as a *thermopile* and is used primarily in measuring small temperature differences. A schematic diagram of a series-connected thermocouple is shown in Fig. M-4.

Thermocouples may also be joined in parallel. In the parallel-connected thermocouple circuit, a mean value of the individual thermocouples is indicated, and it will be the true arithmetic mean if all thermocouple circuits are of equal resistance. A schematic diagram of a parallel-connected thermocouple circuit is shown in Fig. M-5.

The installation of extensive thermocouple equipment requires the services of qualified instrument technicians, and special attention should be given to extension wires, reference junctions, switches, and terminal assemblies.

Opposed thermocouple circuits are sometimes used to obtain a direct reading of a temperature difference between two sets of thermocouples reading two levels of

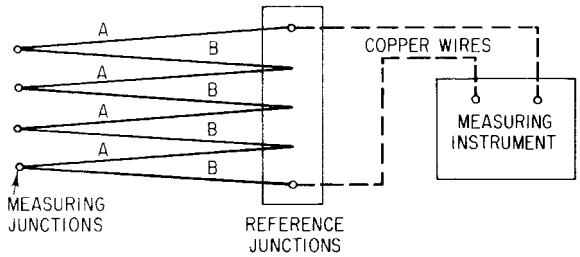


FIG. M-4 Thermocouples connected in series. (Source: Demag Delaval.)

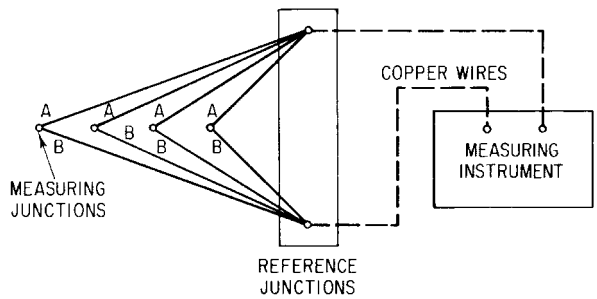


FIG. M-5 Thermocouples connected in parallel. (Source: Demag Delaval.)

temperature. The number of thermocouples in each set is the same. This method is considered to provide the highest degree of accuracy in the measurement of the critical temperature difference.

Filled-system thermometer

A filled-system thermometer (Fig. M-6) is an all-metal assembly consisting of a bulb, a capillary tube, and a Bourdon tube and containing a temperature-responsive fill. Associated with the Bourdon is a mechanical device that is designed to provide an indication or record of temperature.

The sensing element (bulb) contains a fluid that changes in physical characteristics with temperature. This change is communicated to the Bourdon through a capillary tube. The Bourdon provides an essentially linear motion in response to an internally impressed pressure or volume change.

Filled-system thermometers may be separated into two types: those in which the Bourdon responds to volume changes and those that respond to pressure changes. The systems that respond to volume changes are completely filled with mercury or other liquid, and the system that responds to pressure changes is either filled with a gas or partially filled with a volatile liquid.

Bimetallic thermometer

A bimetallic thermometer (Fig. M-7) consists of an indicating or recording device, a sensing element called a bimetallic-thermometer bulb, and a means for operatively connecting the two. Operation depends upon the difference in thermal expansion of two metals. The most common type of bimetallic thermometer used in industrial applications is one in which a strip of composite material is wound in the form of a helix or helices. The composite material consists of dissimilar metals that

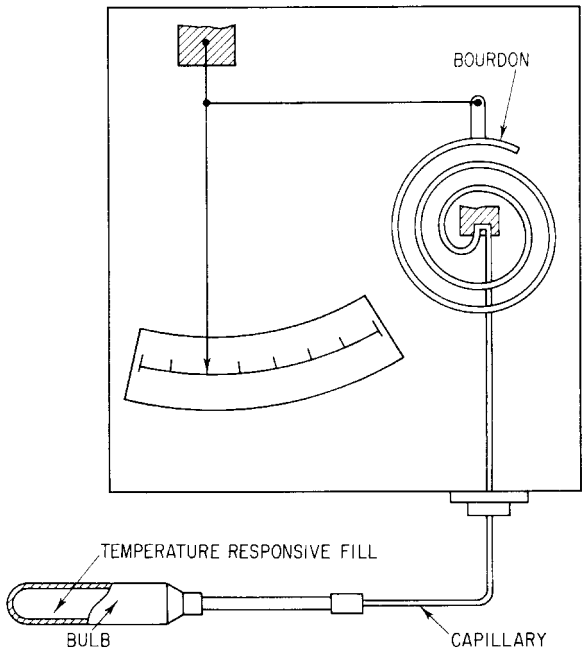


FIG. M-6 Filled-system thermometer. (Source: Demag Delaval.)

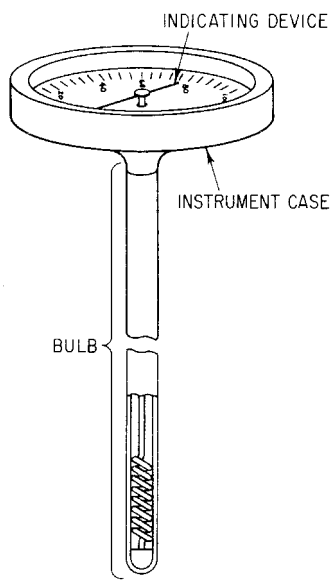


FIG. M-7 Bimetallic thermometer. (Source: Demag Delaval.)

have been fused together to form a laminate. The difference in thermal expansion of the two metals produces a change in curvature of the strip with changes in temperature. The helical construction is used to translate this change in curvature to rotary motion of a shaft connected to the indicating or recording device.

A bimetallic thermometer is a relatively simple and convenient instrument. It comes in industrial and laboratory versions.

Pyrometry

There are two distinct pyrometric instruments, the radiation thermometer and the optical pyrometer, which are described in greater detail in the following subsections. Both pyrometers utilize radiation energy in their operation. Some of the basic laws of radiation transfer of energy will be described briefly.

All bodies above absolute-zero temperature radiate energy. This energy is transmitted as electromagnetic waves. Waves striking the surface of a substance are partially absorbed, partially reflected, and partially transmitted. These portions are measured in terms of absorptivity α, reflectivity ρ, and transmissivity τ, where

$$\alpha + \rho + \tau = 1 \tag{M-4}$$

For an ideal reflector, a condition approached by a highly polished surface, $\rho \rightarrow 1$. Many gases represent substances of high transmissivity, for which $\tau \rightarrow 1$, and a blackbody approaches the ideal absorber, for which $\alpha \rightarrow 1$.

A good absorber is also a good radiator, and it may be concluded that the ideal radiator is one for which the value of α is equal to unity. In referring to radiation as distinguished from absorption, the term *emissivity* ε is used rather than *absorptivity* α. The Stefan-Boltzmann law for the net rate of exchange of energy between two ideal radiators A and B is

$$q = \sigma(T^4{}_A - T^4{}_B) \tag{M-5}$$

where q = radiant-heat transfer, Btu/h·ft^2; σ = Stefan-Boltzmann constant; and T_A, T_B = absolute temperature of two radiators.

If we assume that one of the radiators is a receiver, the Stefan-Boltzmann law makes it possible to measure the temperature of a source by measuring the intensity of the radiation that it emits. This is accomplished in a radiation thermometer.

Wien's law, which is an approximation of Planck's law, states that

$$N_{b\lambda} = C_1 \lambda^{-5} e^{-C_2/\lambda T} \tag{M-6}$$

where $N_{b\lambda}$ = spectral radiance of a blackbody at wavelength λ and temperature T; C_1, C_2 = constants; λ = wavelength of radiant energy; and T = absolute temperature.

The intensity of radiation $N_{b\lambda}$ can be determined by an optical pyrometer at a specific wavelength as a function of temperature, and then it becomes a measure of the temperature of a source.

Radiation thermometer

A radiation thermometer consists of an optical system used to intercept and concentrate a definite portion of the radiation emitted from the body whose temperature is being measured, a temperature-sensitive element, usually a thermocouple or thermopile, and an emf-measuring instrument. A balance is quickly established between the energy absorbed by the receiver and that dissipated by conduction through leads, convection, and emission to surroundings. The receiver equilibrium temperature then becomes the measure of source temperature, with the scale established by calibration. An increase in the temperature of the source is accompanied by an increase in the temperature of the receiver that is proportional to the difference of the fourth powers of the final and initial temperatures of the source.

The radiation thermometer is generally designated as a total-radiation thermometer that utilizes, as an index of the temperature of a body, all the energy

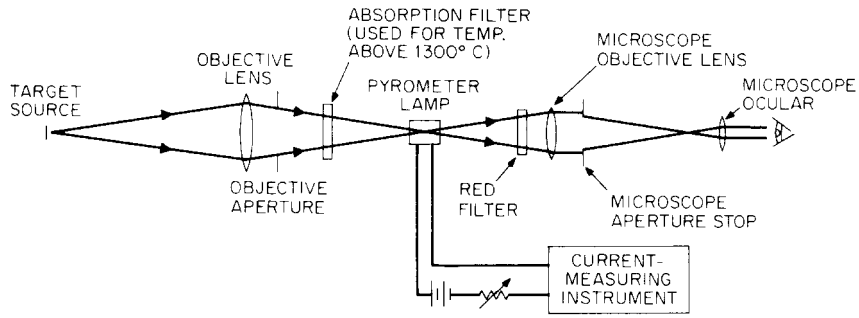


FIG. M-8 Schematic diagram of an optical pyrometer. (Source: Demag Delaval.)

(all wavelengths) per unit area per unit time radiated by the body. Radiation thermometers are classified according to the method of collecting the radiation and focusing it on the receiver: single mirror, double mirror, and lens.

The radiation thermometer can be classified not as a primary laboratory instrument but rather as an industrial instrument. Its practical useful range extends from ambient temperature to 7500°F, although different thermometers must be used to cover this range.

Optical pyrometer

Optical pyrometers use a method of matching as the basis of operation. Generally, a reference temperature is provided in the form of an electrically heated lamp filament, and a measure of temperature is obtained by optically comparing the visual radiation from the filament with that from the unknown source. In principle, the radiation from one of the sources, as viewed by the observer, is adjusted to match that from the other source. Two methods are employed: (1) the current through the filament may be controlled electrically, through a resistance adjustment; or (2) the radiation accepted by the pyrometer from the unknown source may be adjusted optically by means of some absorbing device such as an optical wedge, a polarizing filter, or an iris diaphragm. The two methods are referred to, respectively, as the method using the variable-intensity comparison lamp and the method using the constant-intensity comparison lamp. In both cases the adjustment required is used as the means for temperature readout. Figure M-8 illustrates schematically an arrangement of a variable-intensity pyrometer.

A typical optical pyrometer consists of a power supply and an optical system. The optical system incorporates a telescope, a calibrated lamp, a filter for viewing nearly monochromatic radiation, and an absorption glass filter (see Fig. M-8). The filament of the lamp and the test body are viewed simultaneously. The filament current is adjusted until the filament image disappears in the image of the test body.

Visual optical pyrometers should not be used for the measurement of temperatures below 1400 °F. Automatic optical pyrometers can be used for the measurement of lower temperatures, and they are of great value in the measurement of very high temperatures.

Calibration

To compare or to measure temperature, a temperature scale is necessary. Two ideal temperature scales were proposed: the thermodynamic scale of Kelvin and the ideal-gas scale. The International Committee on Weights and Measures came up with a

more practical temperature scale, the International Practical Temperature Scale of 1968 (IPTS-68), which is based on 11 fixed, reproducible temperature points.

There are two widely used temperature scales in engineering practice. The first, the Celsius scale, derives directly from IPTS-68; it has 100 units (degrees) between the ice point and the steam point of water. The second, the Fahrenheit scale, has 180 units (degrees) between these two fixed temperature points. In the first case the freezing point is marked 0, while in the second case this point is marked 32. The relationship between the two scales is as follows:

$$F = {}^{9}/_{5}\,C + 32, \text{ degrees Fahrenheit}$$

$$C = {}^{5}/_{9}\,(F - 32), \text{ degrees Celsius}$$

Calibration at fixed points is a complex process. Standard platinum resistance thermometers and standard platinum-rhodium-platinum thermocouples are calibrated at fixed points for use as primary standards. It is recommended that calibration be done by the NBS or other qualified laboratory. The narrow-band optical pyrometer is another primary standard; its range over the freezing point of gold is obtained through extrapolation. Ordinary calibration of temperature-measuring instruments is effected by comparison of their readings with those of primary or secondary standards at temperatures other than fixed points. Comparators are used to produce those temperatures.

Secondary standards are liquid-in-glass thermometers and base-metal thermocouples. They are calibrated by comparing them with primary-standard platinum-resistance thermometers or standard platinum-rhodium versus platinum thermocouples at temperatures generated in comparators. These secondary standards are used in turn for the calibration of other devices, such as liquid-in-glass thermometers, bimetallic thermometers, filled-system thermometers, and base-metal thermocouples, in which the highest degree of accuracy is not required. Optical pyrometers as secondary standards are compared with primary-standard optical pyrometers, and they are then used for calibration of regular test pyrometers.

There is ample literature by the American Society of Mechanical Engineers (ASME), the American Society for Testing and Materials, the NBS, and others that deals with calibration methods, specifications for construction and usage of measuring instruments and temperature comparators, and processing of calibration data. It is advisable in each case to have the major components of the system (primary and secondary standards), potentiometers, and Mueller bridges calibrated periodically by the NBS or other qualified laboratory.

Other considerations

The preceding presentation on temperature measurement shows clearly how complex the subject is and what precautions must be taken to obtain a meaningful temperature measurement. The proper use of the right temperature-measuring instruments is very important. Calibration for instrumental errors is mandatory for temperature-sensing devices and other temperature-measurement-system components; periodic checking of the calibration is also very important.

If for reasons of protection of the sensitive temperature-measuring element against corrosive atmosphere or excessive mechanical stress, the use of thermometer wells is prescribed, such wells should be designed and installed with the utmost care to avoid damage and the introduction of additional errors. The *ASME Power Test Codes* should be followed in this respect. The most important precautions in using a thermometer well are to keep the sensing element in

intimate contact with the well and to have the exposed parts of the well as small as possible and insulated from their surroundings.

The nature of heat transfer between the medium, the temperature of which is being measured, and the sensing element and the sources of temperature errors due to conduction, radiation, and aerodynamic heating are described below. The temperature-sensing element indicates its own temperature, which may not be the exact temperature of the fluid in which it is inserted. The indicated temperature is established as a result of heat-flow equilibrium of convective heat transfer between the sensing element and the fluid on one side and heat flow through conduction and radiation between the element and its surroundings on the other side. This applies closely to fluids at rest or to fluids moving with low velocities. The conditions are more complex for fluids moving at higher velocities (corresponding to a Mach number greater than 0.3), in which the aerodynamic heating effect plays a greater part in heat balance.

Conduction error. Conduction error, or immersion error, is caused by temperature gradients between the sensing element and the measuring junction. This error can be minimized by high heat convection between fluid and sensor and low heat conduction between sensor and measuring junction. In the thermocouple this would mean a small diameter, low conductivity, and long immersion length of the wires.

Radiation error. When the sensing element (other than radiation thermometer) is placed so that it can "see" surfaces at a much lower temperature (a sink) or at a much higher temperature (a source), a radiant-heat interchange will result between the two, causing the sensing element to read an erroneous temperature.

Radiation error may be largely eliminated through the proper use of thermal shielding. This consists in placing barriers to thermal radiation around the probe, which prevent the probe from seeing the radiant source or sink, as the case may be. For low-temperature work, such shields may simply be made of sheet metal appropriately formed to provide the necessary protection. At higher temperatures, metal or ceramic sleeves or tubes may be employed. In applications in which gas temperatures are desired, care must be exercised so as not to cause stagnation of flow around the probe.

Measurement of temperature in a rapidly moving gas. When a probe is placed in a stream of gas, the flow will be partially stopped by the presence of the probe. The lost kinetic energy will be converted to heat, which will have some bearing on the indicated temperature. Two "ideal" states may be defined for such a condition. A true state would be that observed by instruments moving with the stream, and a stagnation state would be that obtained if the gas were brought to rest and its kinetic energy completely converted to heat, resulting in a temperature rise. A fixed probe inserted into the moving stream will indicate conditions lying between the two states.

An expression relating stagnation and true temperatures for a moving gas, with adiabatic conditions assumed, may be written as follows:

$$t_s - t_t = \frac{V^2}{2g_c J c_p} \tag{M-7}$$

This relation may also be written as

$$\frac{t_s}{t_t} = 1 + \frac{(k-1)M^2}{2} \tag{M-8}$$

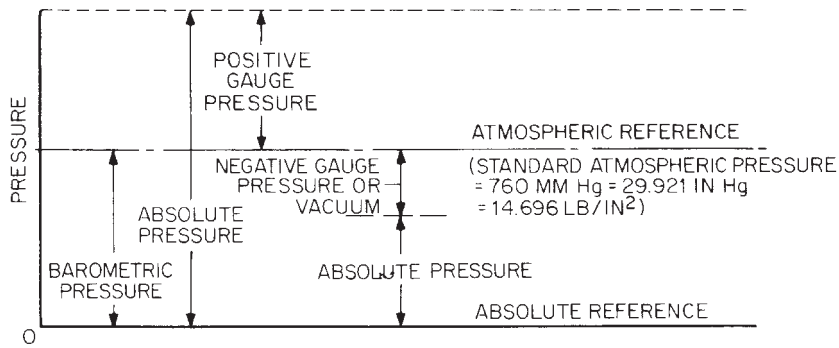


FIG. M-9 Relations between absolute, gauge, and barometric pressure. (Source: Demag Delaval.)

where t_s = stagnation or total temperature, °F; t_t = true or static temperature, °F; V = velocity of flow, ft/s; g_c = gravitational constant, 32.2 ft/s^2; J = mechanical equivalent of heat, ft · lb/Btu; c_p = mean specific heat at constant pressure, Btu/lb · °F; k = ratio of specific heats; and M = Mach number.

A measure of effectiveness of the probe in bringing about kinetic-energy conversion may be expressed by the relation

$$r = \frac{t_i - t_t}{t_s - t_t} \tag{M-9}$$

where t_i = temperature indicated by the probe, °F, and r = recovery factor.

If $r = 1$, the probe would measure the stagnation temperature, and if $r = 0$, it would measure the true temperature.

By combining Eqs. (M-7) and (M-9), the following relationships are obtained:

$$t_t = t_i - \frac{rV^2}{2g_c J c_p} \tag{M-10}$$

or

$$t_s = t_i + \frac{(1-r)V^2}{2g_c J c_p} \tag{M-11}$$

The value of the recovery factor r depends on the type and design of the temperature-measuring probe; it can be anywhere between 0 and 1.0. Often it is specified by the manufacturer for the specific designs of the temperature probes, or it should be determined experimentally. The difference between stagnation and static temperature increases rapidly as the flow Mach number increases. It is important therefore to know the value of the recovery factor in order to get an as accurate as possible evaluation of the temperatures of the moving gas.

Pressure Measurement

General principles and definitions

1. *Pressure* is defined as the force per unit area exerted by a fluid on a containing wall.

2. *Pressure relationships* (Fig. M-9). *Differential pressure* is the difference between any two pressures. *Absolute pressure* is the force per unit area exerted by

a fluid on a containing wall. *Gauge pressure* is the difference between absolute pressure and ambient-atmospheric pressure. *Vacuum pressure* is negative gauge pressure.

3. *Flow-stream pressures*. *Static pressure* is pressure measured perpendicularly to the direction of flow. This is the pressure that one would sense when moving downstream with the fluid. *Total pressure* is pressure in the direction of flow, where pressure as a function of direction is at a maximum. Total pressure would be sensed if the stream were brought to rest isentropically. *Velocity pressure* is the difference between static and total pressure measured at a specific region in the direction of flow. It is called *velocity head* when measured in height of fluid. Velocity pressure is equal to $^1/_2\rho V^2$, where ρ is the fluid density and V is the fluid velocity.

Pressure connections

1. *Sources of errors. Flow errors:* Leakage errors can be eliminated by proper sealing of connections. Errors due to friction, inertia, and lag errors in the gauge piping, encountered in dynamic flow, can be minimized by using short large-diameter connecting tubes. *Turbulence errors:* The static-pressure tap on the wall parallel to the flow should not be too large in order to prevent a disturbance in the flow that would cause an inaccurate static reading; the tap, however, should be large enough to give a proper response. The area surrounding the pressure tap should be smooth to ensure that a burr or other obstruction will not affect the reading. The edge of the hole should be sharp and square. When the pressure is fluctuating, a damping device can be used to improve readability, although the accurate method would be to use a suitable recording instrument and determine the average pressure over a period of time.

2. *Static taps*. Static taps (Fig. M-10*a*) should be at least 5 diameters downstream from symmetrical pipe fittings and 10 diameters downstream from unsymmetrical fittings, according to the *ASME Power Test Codes*. When possible, a Weldolet or pipe coupling should be welded to the outside of the pipe and the hole then drilled through to the main pipe. Since the error increases with velocity pressures, care must be taken in high-velocity areas to ensure sharp, square holes that are as small as possible (down to $^1/_{16}$ in) to keep disturbance and error to a minimum. In low-velocity areas, larger holes should be used to improve dynamic-pressure response and prevent clogging. When flow is nonuniform, several taps should be used along the periphery of the pipe.

3. *Static tubes*. Static tubes (Fig. M-10*b*) are used for measurement of static pressure in a free stream as on a moving plane. Static taps in the wall are preferable, since static tubes disturb the flow, making calibration necessary for accurate measurement. Unless one expects a static-pressure distribution, wall taps should be used.

4. *Impact tubes or pitot tubes*. An impact tube (Fig. M-10*c*) faces directly into the flow, giving a total-pressure reading. Velocity pressure is determined by taking a static reading, preferably along a wall, and taking the difference; impact tubes can be used to get a velocity profile by traversing. Maximum-velocity direction can be determined by rotating the tube.

5. *Piping arrangement*. Connecting piping (Fig. M-10*f*) should be arranged to avoid liquid pockets in gas-filled lines and air pockets in liquid-filled lines. This is accomplished by having gas-filled lines sloping up to the measuring instrument and liquid-filled lines sloping downward to the instrument. Both types should have vents close to the instrument to bleed lines. More vents might be needed if lines must have dips or twists in them. For vacuum pressures an air bleed allowing

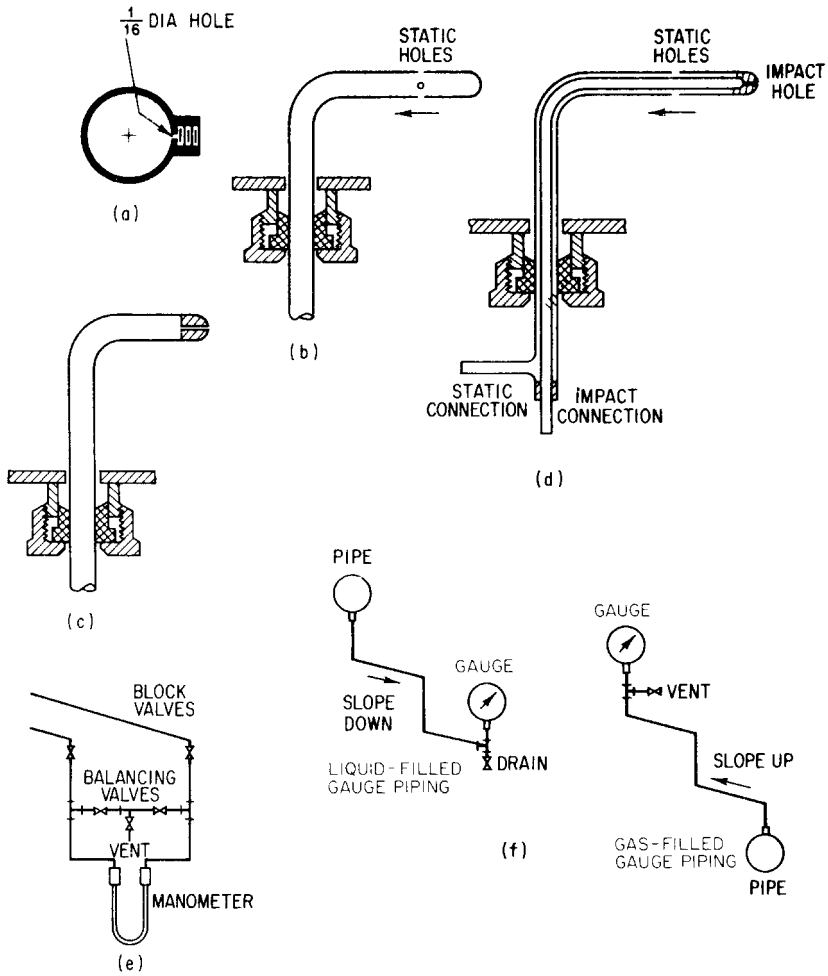


FIG. M-10 (*a*) Static-pressure connection. (*b*) Static tube. (*c*) Impact tube. (*d*) Combination pitot-static tube. (*e*) Cross connection. (*f*) Typical pressure-gauge piping arrangements. (Source: Demag Delaval.)

very small flow should be provided near the instrument to keep lines purged of condensate, etc., between readings. When using a manometer, this can be accomplished by a valve, or a very small hole can be drilled near the top of the manometer, which would be closed or covered when taking a reading. For mechanical or electrical transducers as measuring instruments, the same procedures hold true, but in hot-steam lines it might be necessary to loop the line and fill with water close to the instrument to protect the instrument from the high temperature. For differential measurement the arrangement shown in Fig. M-10*e* should be used to prevent and detect leakage.

6. *Calibration and error analysis*. Pressure measurements are referred to primary standards of pressure, which can be calibrated in terms of mass, length, and time. All pressure-measuring devices have an associated error that must be considered in making a pressure measurement. In a field environment of noise, vibration, moisture, temperature fluctuation, pressure fluctuation, pressure-tap geometry, connecting tubing, etc., other errors or uncertainties must be considered in evaluating the pressure measurement.

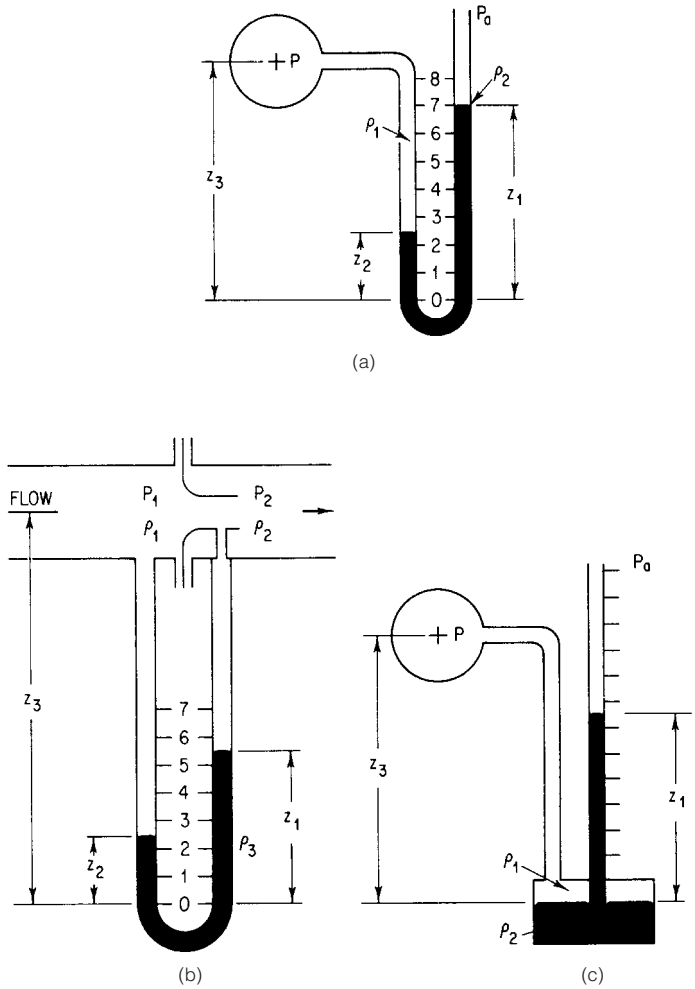


FIG. M-11 Manometer types. (*a*) U-tube manometer, open to the atmosphere. (*b*) Differential U-tube manometer. (*c*) Cistern manometer. (Source: Demag Delaval.)

Liquid-level gauge

1. A manometer measures pressure by balancing it against a column of liquid with a known density and height. Selection of the liquid depends on test conditions; however, the liquid must always be denser than the flowing fluid and immiscible with it. Other factors to consider are the specific gravity, the useful temperature range, the flash point, the viscosity, and the vapor pressure. The basic manometer liquids used are water (specific gravity, 1), mercury (specific gravity, 13.57), red oil (Meriam; specific gravity, 0.827), tetrabromoethylene (specific gravity, 2.95), and carbon tetrachloride (specific gravity, 1.595). Special fluids are also available with specific gravities of 1.20 and 1.75. The manometer fluid used must be kept pure to ensure that the specific gravity remains constant.

2. The basic types of manometers are U-tube and cistern (Fig. M-11).

In the U-tube manometer the pressure on one leg balances the pressure on the other leg. By performing a fluid balance and knowing the density of all fluids and their height, one can calculate the pounds per square inch difference between the two. Often the second leg is open to atmospheric pressure so that the pressure difference represents gauge pressure and must be added to barometric pressure to

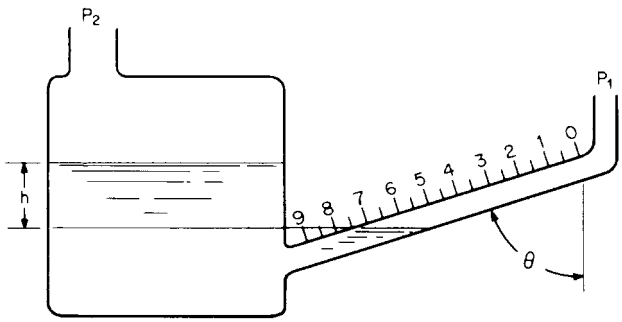


FIG. M-12 Inclined manometer. (Source: Demag Delaval.)

find the total pressure. When the second leg is connected to a pressure other than atmospheric, it is called a *differential pressure* and represents the direct difference between the two pressures.

In the well- or cistern-type manometer one leg has a cross section much larger than that of the other leg. The zero adjustment in the cistern is usually made manually with an adjusting screw. Then the pressure is found by the following formula

$$P = P_a + Z_1 g\rho_2 - Z_3 g\rho_1$$

where g = acceleration due to gravity and ρ = density of liquid.

Special types of manometers sometimes used for more accurate measurement include the inclined manometer, the micromanometer, and U tubes installed with hook gauges, as well as many special types of manometers for vacuum measurement, which will be mentioned later.

3. An inclined manometer (Fig. M-12) is a manometer inclined at an angle with the vertical. Although the vertical displacement is still the same, the movement of liquid along the tube is greater in proportion to the secant of the angle. The common form of inclined manometer is made with a cistern, as shown in Fig. M-12.

The scale can be graduated to take account of the liquid density, inclination, and cistern-level shift so that readings will be in convenient pressure units such as equivalent vertical inches or centimeters of water. A spirit level and leveling screws are usually provided so that the designed angle can be reproduced in installation.

This form of manometer is useful for gas pressures, as for draft gauges. The graduation intervals are commonly 0.01 in of water (0.25 mm of water) with spans up to about 10 in (25 cm).

4. Barometers are a special case of manometers to measure atmospheric pressure. A primary barometer is a U tube with one end open to the atmosphere and the other end connected to a continuously operating vacuum pump.

In many cases a Fortin-type barometer (Fig. M-13) is suitable. In this case the mercury in the well is exposed to the atmosphere with the other end evacuated and sealed. All barometer readings should be corrected for temperature, local gravity, and capillary effect. Atmospheric pressure can also be measured by an aneroid barometer, which is a special type of elastic gauge. It is sometimes used in place of a manometer-type barometer because of the ease of transportation.

Deadweight tester and gauges

1. *Principle, design, and operation testers.* Deadweight testers are the most common instrument for calibrating elastic gauges with pressures in the range of 15 to about 10,000 lb/in^2 or higher.

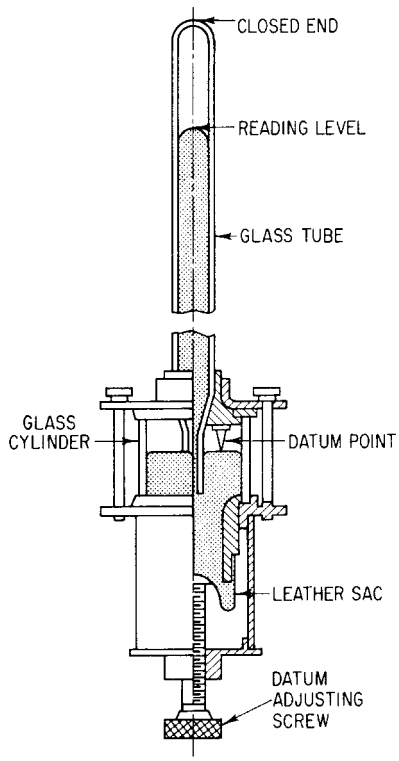


FIG. M-13 Fortin barometer. (Source: Demag Delaval.)

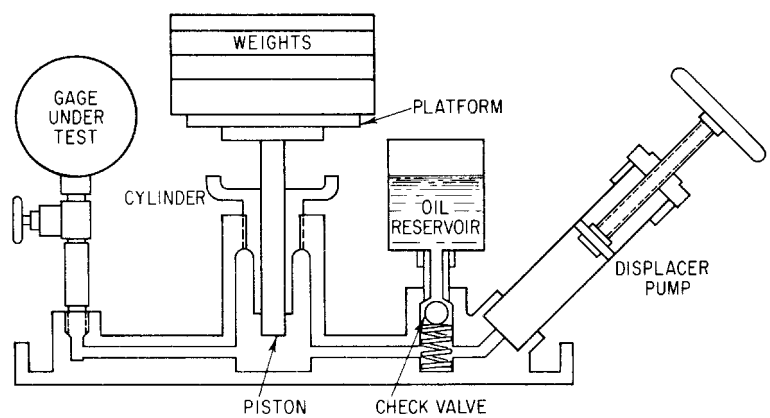


FIG. M-14 Deadweight tester. (Source: Demag Delaval.)

Deadweight testers (Fig. M-14) have a piston riding in a cylinder with a close clearance. The total weight on the piston including that of the platform and the piston itself and any additional weights, divided by the cross-section area of the piston (which is usually an even fraction of an inch such as $^1/_8$ in^2), determine the pressure on the gauge being tested. The piston must be in a vertical position and spinning freely when the measurement is taken. The inertia created by spinning minimizes the viscous drag on the piston by spreading oil around the diameter. Maximum error is usually 0.1 percent of the pressure measured.

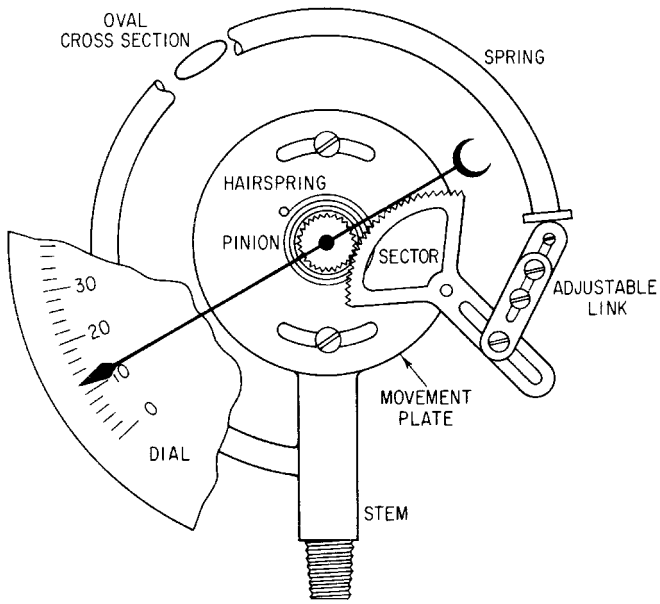


FIG. M-15 Bourdon gauge. (Source: Demag Delaval.)

To operate, put the desired weight on the piston, close the pressure-release valve, and pressurize the tester's fluid with the displacer pump or screw-type ram until the weights are lifted and the piston is floating. Then slowly spin the piston, and take the gauge reading and compare it with the equivalent pressure created by the piston and weights. The gauge reading must then be corrected accordingly.

Special testers include high-pressure, low-pressure, and lever types. For very high pressure (above 10,000 lb/in^2) it is necessary to use a tester that makes adjustments to minimize the leakage and to correct for deformation of the piston and cylinder. Low-pressure testers (0.3 to 50 lb/in^2 are covered) use air as the working fluid for a more accurate measurement. Lever-type testers use a force-amplifying linkage to apply weight to the piston with an inertial wheel on a motor to keep the piston spinning freely.

2. *Deadweight gauges*. Deadweight gauges are mainly used to measure a relatively stable pressure so that it can be maintained. These gauges give very precise measurements but are not practical for a test with a wide range of pressures since many weight changes would be necessary.

3. *Corrections.* Corrections include those necessary for local gravity, weight measurement, effective area, head, and buoyancy adjustments. The head correction is usually the only one necessary when accuracy of $^1/_4$ percent is satisfactory.

Elastic gauges

1. In elastic gauges, an elastic member is caused to stretch or move by a given pressure. The movement is amplified through a linkage and usually is employed to rotate a pointer indicating the pressure reading in relation to atmospheric pressure.

2. Bourdon gauges (Fig. M-15) contain a hollow tube curved in an arc that tends to straighten as internal pressure is applied, moving the linkage and pointer to indicate the pressure reading. Differential as well as compound, vacuum, and straight-pressure Bourdon gauges are available. Differential-pressure gauges have

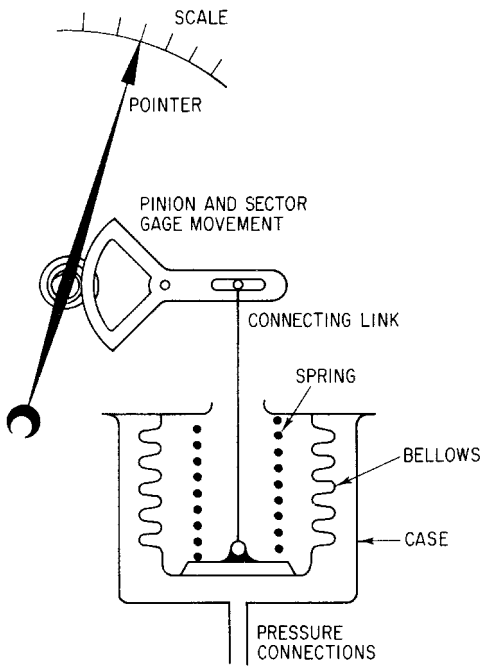


FIG. M-16 Bellows gauge. (Source: Demag Delaval.)

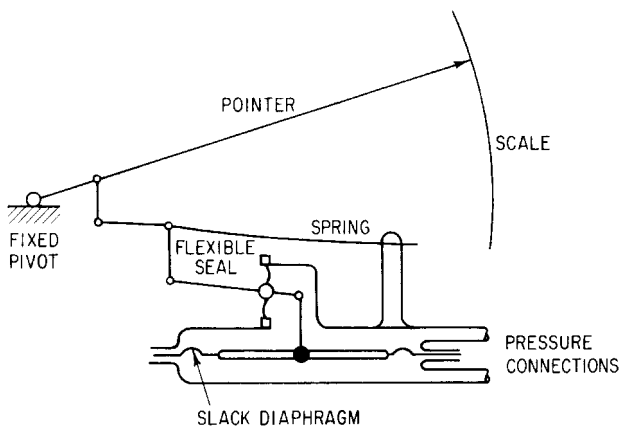


FIG. M-17 Slack-diaphragm gauge. (Source: Demag Delaval.)

either the Bourdon tube enclosed in a seal-pressurized case or two Bourdon gauges, one subtracting from the other. Ranges go from 0 to 15 psig to 0 to 100,000 psig as well as the vacuum range.

3. Bellows gauges (Fig. M-16) have a bellows or elastic chamber expanding to actuate the gauge. They are usually used in low-pressure applications with a maximum reading of about 50 psig.

4. Diaphragm gauges (Fig. M-17) use a flexible diaphragm as the inducer. This type is suitable for ranges from 0 to 1 inHg up to 200 lb/in^2. Variations of this gauge are valuable in special cases in which the process fluid must be kept separate from the gauge, as when the fluid is very hot (up to 1500°F with special modifications) or corrosive, or when the fluid would tend to clog other gauges.

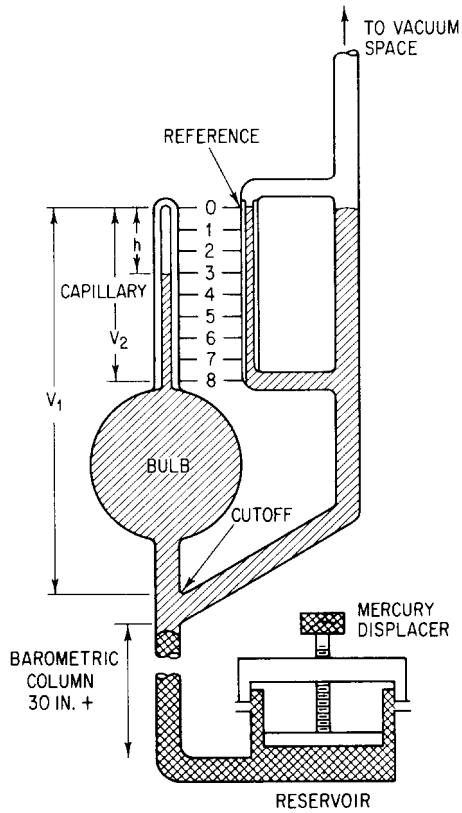


FIG. M-18 McLeod gauge. (Source: Demag Delaval.)

5. All elastic gauges must be calibrated continually to ensure accuracy. Accuracy to 0.5 percent or better of full scale can be obtained.

6. Gauges must be bled for assurance that neither air nor water bubbles are present in the lines. To obtain a gauge reading, first make sure that the linkage is free. This is done with a light tap to the gauge. When damping the gauge needle by closing down on the inlet line, the needle is left fluctuating slightly to indicate that the line is still open.

Special measuring devices

For low-pressure measurement the McLeod gauge (Fig. M-18) is a primary measuring device. The calibration depends only upon dimensional measurements. Other direct-reading pressure gauges, for low pressure, are the mercury micromanometer, the Hickman butyl phthalate manometer, and the consolidated diaphragm comparator. Gauges measuring properties directly convertible to pressure are the thermal-conductivity gauges (thermocouple gauge, Pirani gauge), ionization gauges (Philips-Penning gauge, alphatron gauge), and the molecular-vacuum gauge.

Electric transducers. Devices that convert a pressure into a mechanical analog of that pressure, such as a manometer, which exhibits a difference in the height of a liquid column, were discussed in the preceding subsections. Practical reasons make it difficult to transmit these mechanical signals over large distances, but modern control systems require this capability. Transmitting information over great

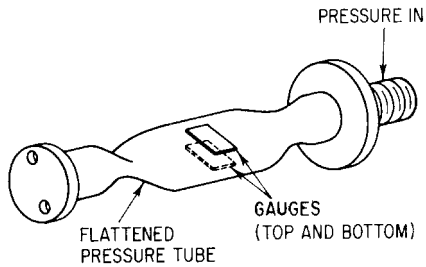


FIG. M-19 Bonded strain gauge. (Source: Demag Delaval.)

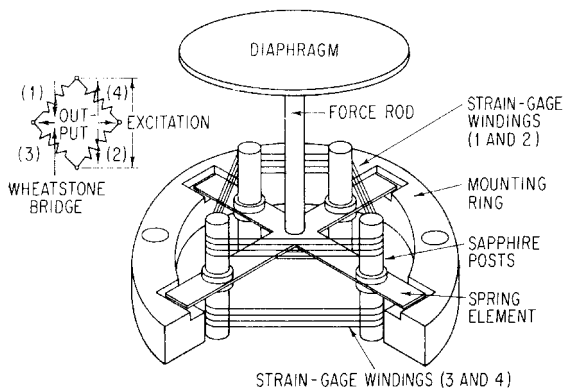


FIG. M-20 Unbonded strain gauge. (Source: Demag Delaval.)

distances is easily accomplished by electronic instrumentation. Transmission of signals representing measured pressure can be accomplished by varying an electric current through wires to the remote location.

The device used to obtain an electronic signal that is related to a pressure is an electric transducer. An electric transducer consists of the following:

1. *Sensing element*. A device that receives a pressure signal and converts it to a force or displacement.
2. *Transmitter*. A device that contains a sensing element, detects the force or displacement in the sensing element, and sends an electric signal (related to the force or displacement) to a receiver.

Implicit in the use of an electric transducer is a receiver that detects the electric signal and indicates the pressure.

Many of these transducers measure the resistance change of a wire or strain gauge deformed by pressure. These instruments can be calibrated to measure pressure directly. There are two basic ways of mounting these gauges. With bonded strain gauges (Fig. M-19) they are usually mounted on a diaphragm or tube that will deform as pressure is applied, changing the resistance of the gauge. With unbonded strain gauges (Fig. M-20) a thin wire is usually wrapped around a sensing element that deforms and stretches the wire, changing its resistance, which can also be converted to pressure. Another special type of resistance gauge for high-pressure reading is the bulk-modulus pressure gauge (Fig. M-21), which uses direct pressures on a loosely wound coil of fine wire to get a resistance change. The sensing mechanism is separated from the process fluid by a bellows. Other specialized types use differences in inductance or capacitance and correlate them to pressure.

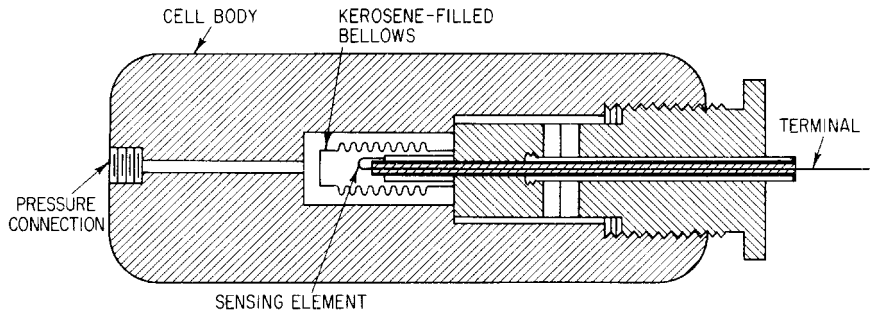


FIG. M-21 Section through a bulk-modulus pressure gauge. (Source: Demag Delaval.)

Flow Measurement

General

The three most extensively used types of flow-metering devices are the thin-plate square-edged orifice, the flow nozzle, and the venturi tube. They are differential-head instruments and require secondary elements for measurement of the differential pressure produced by the primary element. The Supplement to *ASME Power Test Codes: Instruments and Apparatus*, describes construction of the above primary flow-measuring elements and their installation as well as installation of the secondary elements. The method of flow measurement, the equations for flow computation, and the limitations and accuracy of measurements are discussed. Diagrams and tables showing the necessary flow coefficients as a function of Reynolds number and diameter ratio β are included in the standards. Diagrams of the expansion factor for compressible fluids are given.

Some characteristic features of various types of primary elements are listed in the following:

- *Orifice*. Simple, inexpensive, well-established coefficient of discharge, high head loss, low capacity for given pipe size, danger of suspended-matter accumulation; requires careful installation of pressure connections.
- *Flow nozzle*. High capacity, more expensive, loss comparable with that of the orifice; requires careful installation of pressure connections.
- *Venturi tube*. High capacity, low head loss, most expensive, greater weight and size; has integral pressure connection.

Nomenclature

a = throat area of primary element, in^2
C = coefficient of discharge
d = throat diameter of primary element, in
D = pipe diameter, in
E = $1/\sqrt{1-\beta^4}$, velocity-of-approach factor
F_a = thermal-expansion factor
h = manometer differential pressure, in
h_w = manometer differential pressure, in H_2O at 68°F
k = ratio of specific heats
K = CE = combined-flow coefficient for orifices, velocity-of-approach factor included
n = numerical factor dependent upon units used
q = capacity of flow, gal/min

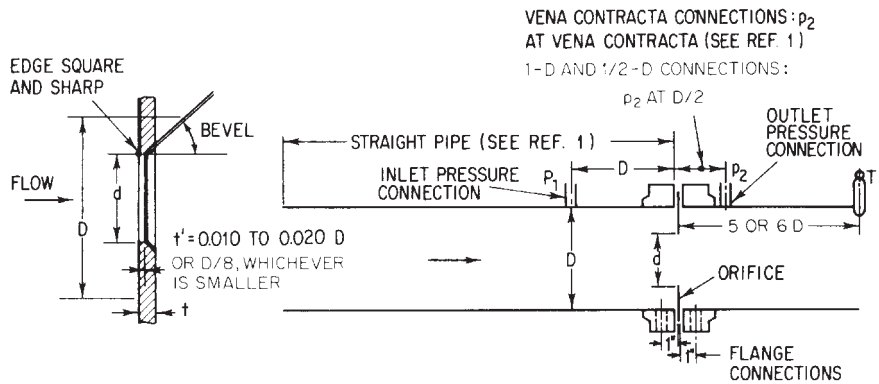


FIG. M-22 Orifice construction and installation. (Source: Demag Delaval.)

Q_i = capacity of flow, ft^3/min, at conditions i
R_d = Reynolds number based on d
$r = \dfrac{P_2}{P_1}$ = pressure ratio across flow nozzle, where P_1 and P_2 are absolute pressures
M = rate of flow, lb/s
M_h = rate of flow, lb/h
M_m = rate of flow, lb/min
Y = net-expansion factor for square-edged orifices
Y_a = adiabatic-expansion factor for flow nozzles and venturi tubes
$\beta = \dfrac{d}{D}$ = diameter ratio
ρ = specific weight of flowing fluid at inlet side of primary element, lb/ft^3
ρ_i = specific weight of flowing fluid at conditions i

Primary-element construction and installation

The primary element may be installed within a continuous section of pipe flowing full or at the inlet or exit of a pipe or a plenum chamber. Orifice and venturi tube are installed within the pipe in a closed-loop test. The flow nozzle may be installed within, at inlet, or at outlet of the pipe.

It is normal practice to use a venturi tube installed within a continuous section of pipe in pump-acceptance tests and a flow nozzle at the exit of the discharge pipe in compressor-acceptance tests. More closed-loop testing has recently been required in compressor testing. Industry normally uses the nozzle configuration shown in Fig. M-26 with a closed loop. The construction of the primary elements and examples of their installation are given in the following paragraphs.

Orifice (Fig. M-22). The recommended diameter ratio $\beta = d/D$ is from 0.20 to 0.75. The thickness of the orifice plate shall be not less than shown in Table M-5.

Three types of pressure connections may be used: vena contracta taps, 1-D and $^1/_2$-D taps, and flange taps. Appropriate discharge coefficients have to be used in each case.

Flow nozzle. The flow nozzle should be of either the low- or the high-ratio long-radius type as shown in Fig. M-23. The diameter ratio β should be from 0.15 to 0.80, although *ASME Power Test Codes: Compressors and Exhausters* (page 22) recommends the β range between 0.40 and 0.60.

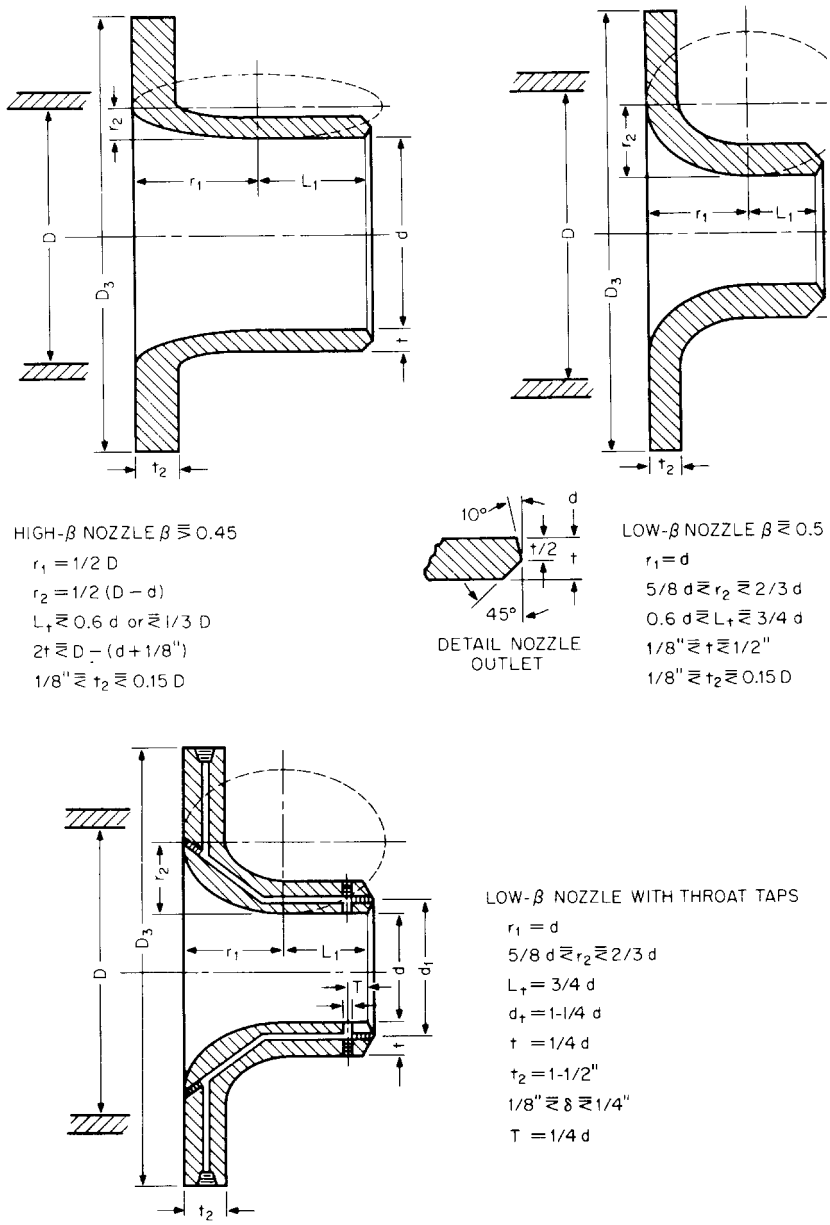


FIG. M-23 Recommended proportions of ASME long-radius flow nozzles. (Source: Demag Delaval.)

Different test arrangements for the measurement of subcritical flow with the flow nozzle are shown in Figs. M-24 through M-26.

Venturi tube. The proportions of the standard-form (Herschel-type) venturi tube and its installation are shown in Fig. M-27. The diameter ratio $\beta = d/D$ should be between 0.4 and 0.75 for the best results. Special forms (venturi-nozzle tube for high-pressure feedwater application, venturi-insert nozzle) may be used. They need individual calibration, however, while graphs may be used for the standard form.

TABLE M-5 Minimum Recommended Thicknesses of Orifice Plates

Differential Pressure, in H_2O	Internal Diameter of Pipe, in				
	3 and Less	6	10	20	30
		$\beta < 0.5$			
<1000	1/8	1/8	3/16	3/8	1/2*
<200	1/8	1/8	1/8	1/4	3/8
<100	1/8	1/8	1/8	1/4	3/8
		$\beta > 0.5$			
<1000	1/8	1/8	3/16	3/8	1/2
<200	1/8	1/8	1/8	3/16	3/8
<100	1/8	1/8	1/8	3/16	1/4

* For 1/2-in plate in 30-in line, maximum differential = 500 in.

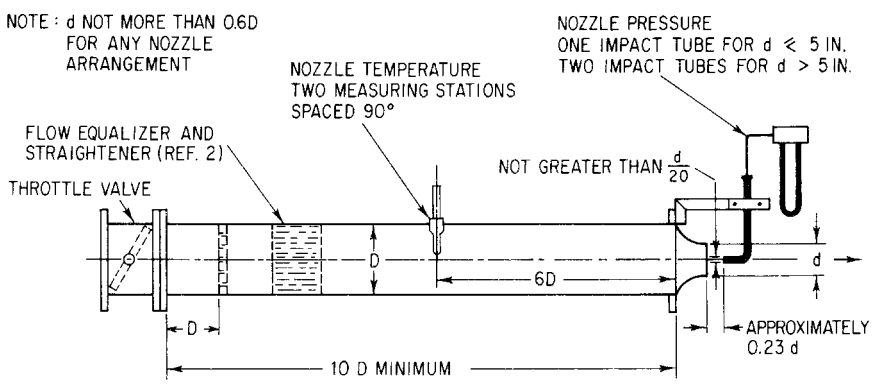


FIG. M-24 Test arrangement *A*. (Source: Demag Delaval.)

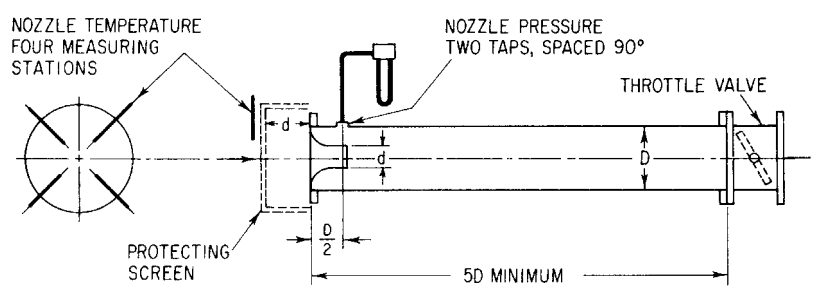


FIG. M-25 Test arrangement *B*. (Source: Demag Delaval.)

Calculation of flow rates

The basic relationships for flow computation were obtained from PTC 19.5, 4-1959, Supplement to *ASME Power Test Codes*, "Flow Measurement of Quantity of Materials." The nomenclature in these relationships was, however, adapted to the more recent terminology of Interim Supplement 19.5, *Fluid Meters*.

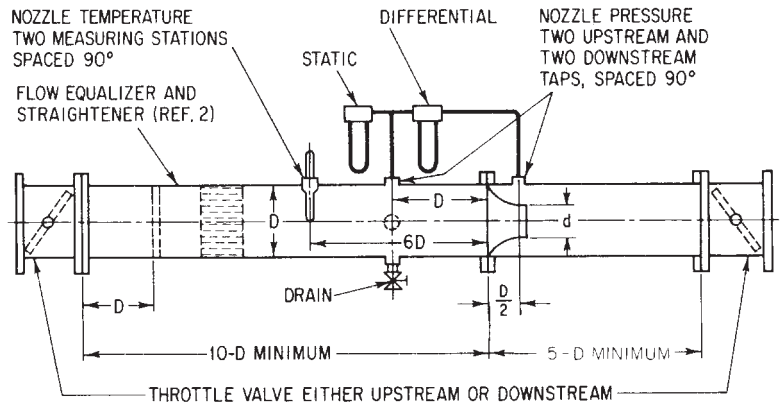


FIG. M-26 Test arrangement *C*. (Source: Demag Delaval.)

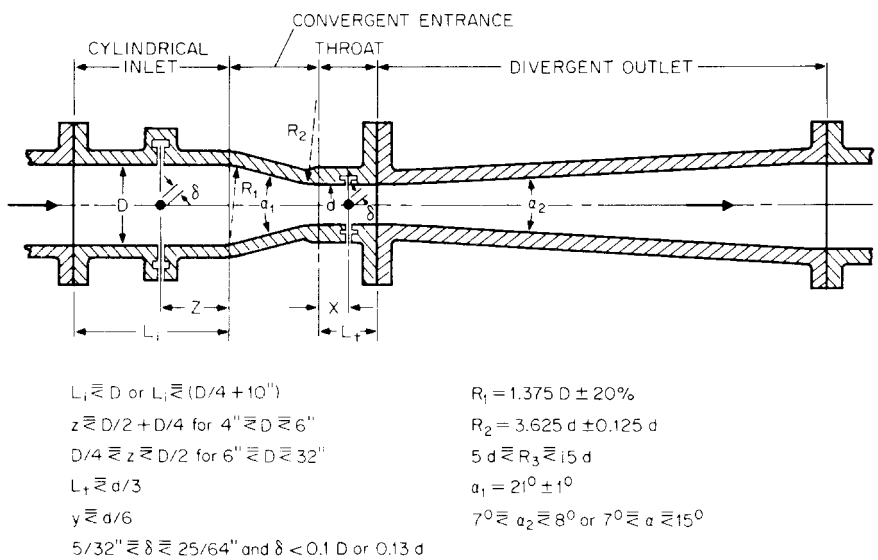


FIG. M-27 Dimensional proportions of classical (Herschel) venturi tubes with a rough-cast convergent-inlet cone. (Source: Demag Delaval.)

Incompressible fluids. The flow of any liquid through an orifice, flow nozzle, or venturi tube is determined by the following equation:

$$M = \frac{CaE_a n\rho}{\sqrt{1-\beta^4}}\sqrt{2gh} \tag{M-12}$$

With the units in American practice, Eq. (M-12) is written

$$M_h = 359CEd^2F_a\sqrt{h_w\rho} \tag{M-13}$$

Compressible fluids. To compensate for the change in specific weight as the fluid passes through the primary element, the equation must be modified by the expansion factor Y for orifices or Y_a for nozzles.

Orifices

$$M_h = 359CEd^2F_aY\sqrt{h_w\rho} \tag{M-14}$$

Flow nozzles and venturi tubes

$$M_h = 359CEd^2F_aY_a\sqrt{h_w\rho} \tag{M-15}$$

Quite often the flow coefficient $K = CE$ is specified for use in Eqs. (M-13) and (M-14) for orifices.

Delaval applications. With particular reference to the methods used at the information source, the above equations assume the forms shown next.

Venturi tube. For measurement of the flow rate of water at 68°F with a *venturi tube* with mercury used as the manometer liquid and a value of 0.984 used for the flow coefficient, Eq. (M-13) becomes

$$q = 19.79\frac{d^2}{\sqrt{1-\beta^4}}\sqrt{h_0} \tag{M-16}$$

where h_0 = manometer pressure, inHg at 68°F

If water at a temperature other than 68°F is metered, a temperature correction should be made.

The coefficient of 0.984 is applicable for β values from 0.3- to 0.75-in pipes of 4 to 32 in, provided the Reynolds number is greater than 200,000. The *Hydraulic Institute Standards* require a minimum velocity of 20 ft/s in the throat for the rated flow of the pump.

Flow nozzle. For measurement of airflow with *flow-nozzle* arrangement *A* (see Fig. M-24),

$$M_m = 5.983CEd^2Y_a\sqrt{h_w\rho} \tag{M-17}$$

and

$$Q_i = \frac{M_m}{\rho_i} \tag{M-18}$$

where

$$Y_a = r^{1/k}\left[\frac{k}{k-1}\frac{1-r^{(k-1)/k}}{1-r}\right]^{1/2} = \text{adiabatic-expansion factor} \tag{M-19}$$

The formula applies to subcritical flow with pressure ratio P_2/P_1 preferably greater than 0.7.

Unless the nozzle in this arrangement has been individually calibrated, the value of the flow coefficient should be taken as 0.99, provided that the throat Reynolds number exceeds 200,000. For nozzles within the pipe and using pipe taps, the coefficient of discharge for pipe Reynolds numbers above 100,000 varies from 0.980 to 0.993, depending on β. Exact coefficients can also be obtained from Tables II–III-5 in ASME Supplement 19.5 on *Fluid Meters*.

For a critical flow measurement, an arrangement similar to that of Fig. M-24 with static-pressure taps 1 *D* upstream from the pipe exit but without an impact tube is used. Rules of the *ASME Power Test Codes: Compressors and Exhausters* should be used for flow computation.

Table M-6 gives approximate flow rates for different nozzle sizes. It is useful for air tests with flow-nozzle arrangements *A* and *B*, when using a long-radius low-β-series nozzle (see Fig. M-23).

TABLE M-6 Long-Radius Low-Ratio Nozzle

	Approximate Flow Rates, ft^3/min	
d	10 in H_2O	40 in H_2O
2.000	250	500
2.500	400	800
3.000	575	1,150
4.000	1,000	2,000
5.000	1,600	3,200
6.000	2,250	4,500
8.000	4,000	8,000
10.000	6,250	12,500
12.000	9,000	18,000
18.000	20,000	40,000
24.000	36,000	72,000

SOURCE: *ASME Power Test Codes*, PTC 10-1965, reaffirmed 1973, *Compressors and Exhausters*.

TABLE M-7 Values of Discharge Coefficient *C* for 8-in Pipe (Velocity-of-Approach Factor Not Included)

R_d / β	Reynolds Number					
	14,000	25,000	50,000	100,000	500,000	1,000,000
	Flange Taps					
0.250	0.6190	0.6091	0.6028	0.5997	0.5971	0.5968
0.500	0.6443	0.6259	0.6142	0.6084	0.6037	0.6032
0.750	. . . . .	0.6749	0.6351	0.6153	0.5993	0.5973
	Taps at 1 D and $\frac{1}{2}$ D					
0.250	0.6036	0.6012	0.5990	0.5975	0.5955	0.5950
0.500	0.6150	0.6112	0.6079	0.6055	0.6024	0.6017
0.750	. . . . .	0.6337	0.6254	0.6196	0.6118	0.6099
	Vena Contracta Taps					
0.250	0.6071	0.6035	0.6003	0.5981	0.5952	0.5945
0.500	0.6185	0.6136	0.6094	0.6064	0.6024	0.6014
0.750	. . . . .	0.6318	0.6240	0.6185	0.6112	0.6095

Orifice. Equation (M-13) is used for measuring the flow rate of water with orifice and Eq. (M-14) for measuring the rate of flow of air. Interim Supplement 19.5 on *Fluid Meters* gives the tables of the values of discharge coefficient C for various pipe diameters as a function of diameter ratio β and pipe Reynolds number R_d based on orifice diameter d. Excerpts from the tables in the Supplement are given in Table M-7 for the three types of pressure-tap locations. The values below the stepped line are extrapolations and are subject to larger tolerance as given in Table II–V-1 in the Supplement.

References and Additional Reading

Temperature

1. *ASME Power Test Codes*, part 3, PTC 19.3-1974, *Temperature Measurement*.
2. Benedict, R. P., *Fundamentals of Temperature, Pressure, and Flow Measurements*, 2d ed., John Wiley & Sons, Inc., New York, 1977.

Pressure

1. *Meriam Standard Indicating Fluid Bulletin*, IM-11.
2. *ASME Power Test Codes: Instrument and Apparatus*, part 2, *Pressure Measurement*, American Society of Mechanical Engineers, New York, 1964.
3. Beckwith, T. G., and N. Lewis Buck, *Mechanical Measurements*, Addison-Wesley Publishing Company, Inc., Reading, Mass., 1961.
4. Benedict, R. P., *Fundamentals of Temperature, Pressure, and Flow Measurements*, 2d ed., John Wiley & Sons, Inc., New York, 1977.
5. Consolidated Electrodynamics, *Transducer*, General Catalog Bulletin 1322, December 1965.

Flow

1. *Fluid Meters*, part II, *Application*, Interim Supplement 19.5, *ASME Power Test Codes: Instruments and Apparatus*, 6th ed., 1971, pp. 179–256.
2. *ASME Power Test Codes*, PTC 10-1965, reaffirmed 1973, *Compressors and Exhausters*.
3. Supplement to *ASME Power Test Codes: Instruments and Apparatus*, PTC 19.5, 4-1959, chap. 4, "Flow Measurement of Quantity of Materials."
4. *Hydraulic Institute Standards*, 13th ed., 1975, pp. 64–68.

Metallurgy; Metallurgical Repair; Metallurgical Refurbishment

This is another of those fields that merits its own set of handbooks: one each for welding, brazing, coatings, laser repair work, and so on. What follows is just enough information to give the process engineers some idea of how much might be possible in this field, including when they could question component lives as specified by an original equipment manufacturer (OEM) and ask for improvements in their life-cycle assessments (LCAs) (*see* Life-Cycle Assessment). Process engineers can also use the material here for direction on how to proceed with questions to ask a metallurgical repair facility. This is an area that offers considerable room for cost economics optimization. The cost of a standard overhaul for a 20,000-hp gas turbine can easily top $1,000,000, especially if the customer does not maintain a parts pool (storage of critical parts) and recondition or repair parts as much as possible. Over 50 percent of repair costs can be saved with careful and correct use of modern metallurgical repair technology.

At one point in the gas turbine's history, a keen demarcation existed between industrial (be they for power generation or mechanical drive) and aircraft gas turbines. That division gets very hazy with some models, particularly those of manufacturers who make both aviation and land-based machines, such as General Electric and Rolls Royce. Metallurgy becomes particularly quite similar between, for instance, a GE LM2500 (aeroderivative) and a GE Frame 9E (industrial) and a GE CF6-80C2 (aircraft). Rolls Royce makes land-based Trents for power generation and aviation Trents and so forth. While there are differences, there are more important similarities, metallurgically speaking. Typical aeroderivative engines are illustrated in Figs. M-28 (Rolls Royce Avon) and M-29 (GE LM2500). The use of cooling passages (technology transferred from aircraft to aeroderivative engines) helps raise the turbines' TITs (see Fig. M-30).

There is also a sort of hybrid turbine category with manufacturers, such as Siemens, that are officially a land-based machinery manufacturer, but have a technology-sharing agreement with Pratt and Whitney. Pratt, which manufactures aviation turbines as well, also uses its aviation technology where the company believes it might benefit users (and economize on design costs) in their land-based models. The consequences for repair technology are that overhaul shops acquire expertise and equipment at a far higher level of sophistication than they did only 10 years ago. Also, now most refurbishment facilities do both aviation and land-based work to optimize the size of their client base.

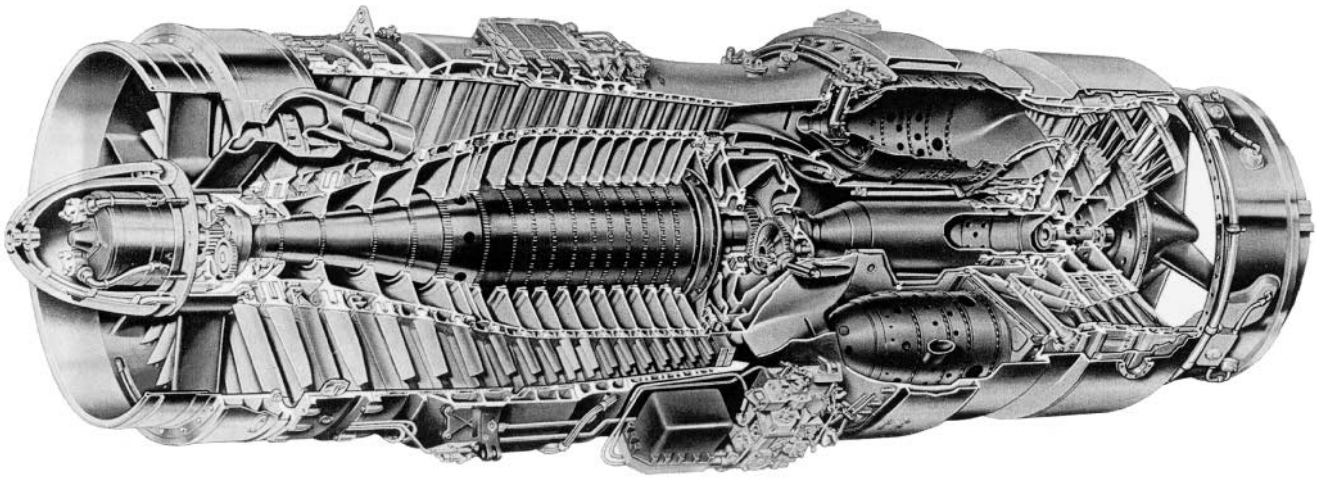


FIG. M-28 Rolls Royce avon aeroderivative gas generator (1535). (Source: Rolls Royce.)

The implications for turbine maintenance are that if the shops use their new technology well, the longevity of parts can be extended, as can time between overhauls. Properly used, sophisticated repair technology can be used to promote the on-condition philosophy that end users are learning to value increasingly. This fact is best illustrated by looking at the nature of some of these repair processes.

The new consciousness regarding environmentally friendly processes sometimes adds initial capital cost, and end users have to bear their share. However, their overall costs per fired hour drop over a longer period, as better repair techniques mean that the cost of "repeat" repairs (done when a process doesn't work right the first time) drops sharply.

What follows is a summary of some of the most basic techniques available.

Repair/Refurbishment Processes*

Brazing

Activated diffusion healing (ADH): This involves manipulating the mix powdered alloys and repair process temperature.

Partitioned allow component healing (PACH): This uses higher temperatures than ADH and is used on single crystal and directionally solidified materials.

Cleaning and stripping

"Environmentally friendly" is a key term for this stage of repair, which formerly had problems with carcinogen content and other types of hazardous chemical cleaners.

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* Source: Liburdi Engineering, Canada, or as stated.

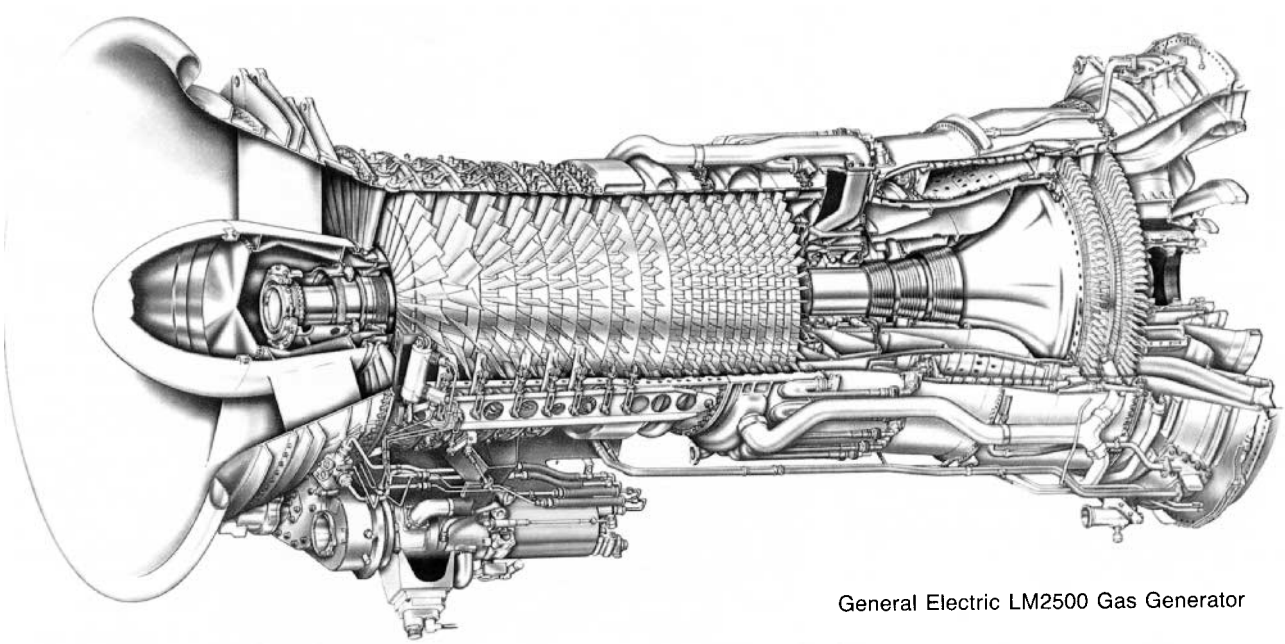


FIG. M-29 General Electric LM2500 gas generator. (Source: General Electric.)

DESIGN FEATURES

Compressor
Type: Axial
No. Stages: 17
Pressure Ratio: 9 to 1
Air Flow: 76 kg per sec ec
Design Speed: 7,550 rpm

Combustor
Type: Cannular
No. Combustors: 8
Ignition: High energy igniters
No. Igniters: 2
No. Fuel Nozzles: 8

Compressor Turbine
Type: Impulse/Reaction
No. Stages: 3
Cooling: 1st & 2nd vanes
1st blades

Bearings
Journal: Roller and ball
Thrust: Ball
Starter
Basic: External drive, motor supplied by packager
Optional: G.G. mounted electric motor

Control System
Supplied by packager

Dimensions
Dry Weight: 1,600 kg
Volume: 3.5 × 1 × 1 m

MATERIALS OF CONSTRUCTION

Compressor
Casing, Front . . . Aluminum Liner . . . Nimonic 263
Casing, Rear . . . Carbon steel
Rotor Blades Stage 00 . . . Titanium alloy
Stage 3–15 . . . Titanium alloy
Stage 0–2 . . . Aluminum alloy
Stator Vanes Stage 00 . . . 12% Cr steel
Stage 5–15 . . . 12% Cr steel
Stage 0–4 . . . Aluminum alloy
Discs . . . Ferritic heat-resistant steel

Combustor
Casing . . . Medium carbon steel

Compressor Turbine
Discs . . . Ferritic heat-resistant steel
1st Vane . . . X-40
2nd & 3rd Vane . . . Nimonic 242
1st Blade . . . Inco 738
2nd Blade . . . Nimonic 115
3rd Blade . . . Nimonic 105

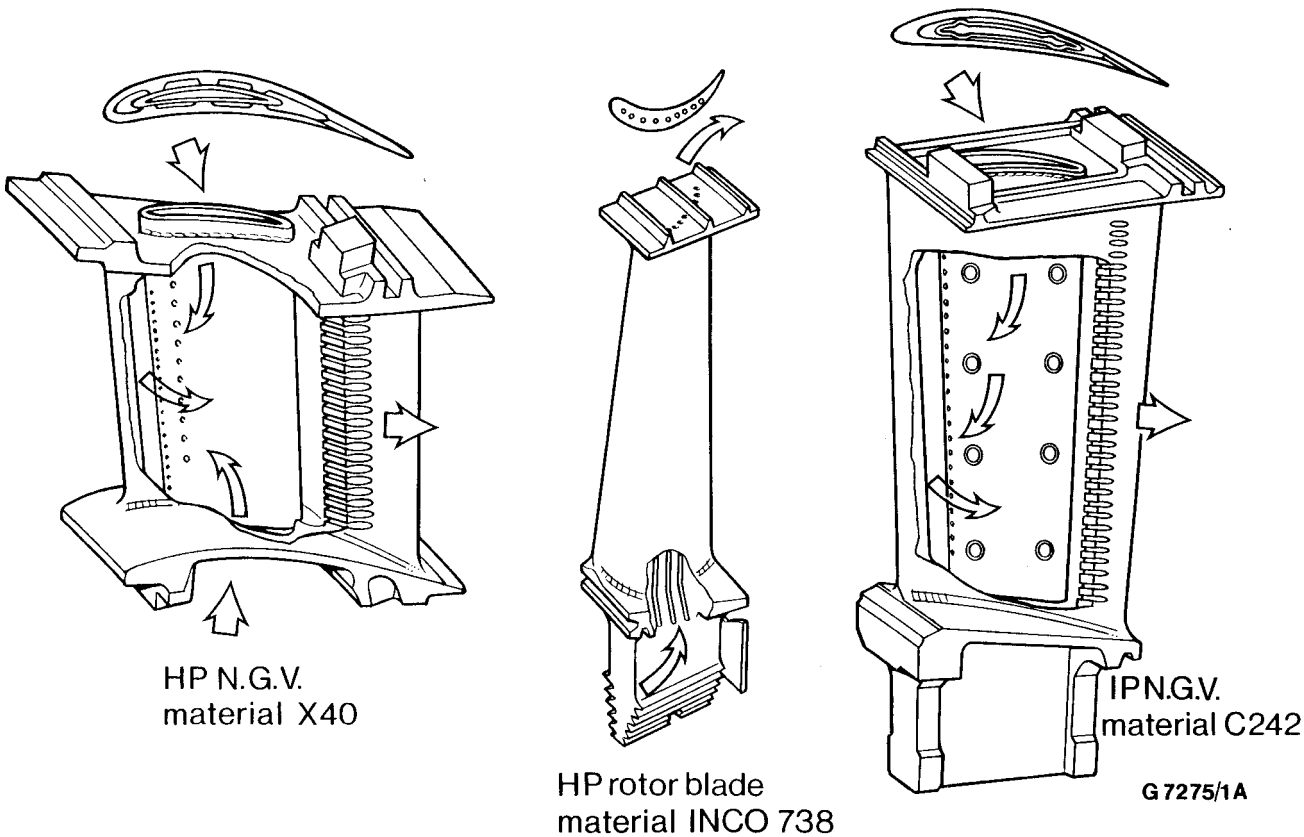


FIG. M-30 Avon Mk. 1535 turbine cooling air. HP = high pressure, N.G.V. = inlet guide vane, IP = intermediate pressure. (Source: Rolls Royce.)

- Plastic cleaners: Clean, recyclable physical abrasives strip contaminants without damaging the parent metal. Note that shot peening ("old" technology) did, especially if the shot was metallic. Glass-bead peening was less damaging, but both "old" peen processes inserted some amount of temporary residual stress, desirable or otherwise. New metallurgy and more sophisticated repair processes can replace conventional peening with beneficial effects on parts' integrity and longevity.
- Autoclaving: Parts are subjected to high pressures and temperatures for superior cleaning.
- Fluoride ion process: This process removes aluminum oxides from parts' surfaces.
- Laser cleaning: Lasers are very accurate and can remove coating or paint layers with great precision and relatively little temperature increase.
- Water jet: This process is also very precise for removing items such as coatings. It can also cut through 10 cm of steel if necessary.

Coatings

Typical examples include:

- Physical vapor deposition (PVD): This is a ceramic coating process and is not required except in the case of high turbine-inlet temperature (TIT) designs. It requires a vacuum.

- Air plasma: Like PVD, except a vacuum is not required.
- Thermally densified coatings (TDC). Generally used to protect shrouds. PVD and TDC are currently used mainly on aircraft engine components, although that will change as OEMs increase the market share of aeroderivatives in land-based use.
- High-velocity oxygen fuel (HVOF): This can be used on bearing surfaces and has replaced hazardous chromates.
- Vapor coatings: Platinum bond coats that provide surfaces that can meet aircraft airfoil and therefore land-based standards. These are nonenvironmental polluting processes.

Machining

Mills, lathes, and grinders have been the conventional workhorses for this process for decades. They are slow and can build up heat in the component if mismanaged. They often need expensive computer numerical control (CNC) to ensure adequate accuracy.

- Water jet: See the same process under Cleaning above.
- Laser machining: See the same process under Cleaning. The time saved is enormous. A component that might have taken 2 to 3 days to machine conventionally can be done with laser technology in half an hour. The initial cost of a laser machining rig is high and end users will bear some part of the initial capital cost, but with the cost of being off-line while they wait for their parts, they generally don't care about any increase in repair cost. There may be none once the time factor is considered.
- Laser drilling: The most accurate kind of drilling, generally used for manufacture of cooling holes on internally cooled turbine blades. However, it is accurate enough that it can be used in overhaul to "redrill" (open up clogged holes) in the same components.
- Electrical discharge machining (EDM): Somewhat older technology than laser drilling and much cheaper, but still fairly accurate for processes such as drilling small holes in components.

Nondestructive test/inspection (NDT/NDI)

Radiographic inspection. Radiographic inspection uses radiation such as X rays or gamma rays to penetrate the material and produce an image of the component on photographic film that can be evaluated and permanently recorded. The exposure time is carefully controlled to produce an image or silhouette of the component that varies in density or shading according to the amount of radiation that reaches each area of the film. When defects are present, the amount of radiation reaching the film is altered resulting in a change in film density directly below the defect location. This enables the radiographer to determine the location and size of the defect. However, narrow defects that are oriented perpendicular to the radiation beam will not be detected because they do not significantly alter the beam transmission.

Turbine manufacturers generally rely on radiographic inspection to control the internal quality of critical castings and welds. Radiography can locate internal defects such as gas porosity, nonmetallic inclusions, cracks, hot tears, shrinkage cavities, lack of fusion, and core shifts. Some of the components inspected include precision cast blades, vane segments, structural welds, and large housing castings.

Ultrasonic inspection. Ultrasonic inspection relies on the use of sound waves to locate discontinuities in a manner similar to radar. The sound waves are generated by a piezoelectric crystal and transmitted through a coupling medium such as oil or water into the component being tested. The high-frequency sound waves penetrate through the material and behave in a manner similar to light waves. When the beam strikes a defect or the opposite wall of the component, the waves are reflected back and re-collected by the crystal. The sound energy is transmitted back into an electrical signal and the information displayed as impulses on a cathode ray screen. The operator must use his skill and knowledge to properly calibrate and interpret the signals and report on the approximate size and location of any defects.

Ultrasonic inspection is capable of locating both internal and surface-connected defects; however, it will not detect narrow cracks oriented parallel to the beam. Turbine manufacturers employ ultrasonics to qualify the internal soundness of critical forgings such as shafts and discs and to determine wall thickness for cooling passages.

Magnetic particle inspection. Magnetic particle inspection is used to locate surface defects in ferromagnetic materials such as steels. A magnetic field is established in the component by either passing electricity directly through prongs or indirectly by wrapping cables around the part. When the lines of force are interrupted by a surface crack, a magnetic leakage field is produced that will attract magnetic particles. Defects can be located by spreading finely divided ferromagnetic material (either dry or in a liquid fluorescent medium) over the surface of the component and observing where the particles are attracted and held in place. The particles form an outline of the defect revealing its relative size and shape. Defects are best resolved when they are oriented at 90° to the direction of the induced magnetic field and, therefore, components should be inspected in at least two directions. The parts must be demagnetized after inspection.

Magnetic particle inspection is used extensively in the turbine industry to inspect forgings, castings, and welds for surface or near surface defects such as cracks, laps, seams, and grinding marks. Components inspected include steel discs, shafts, compressor blades, and vanes and welds on casings.

Eddy current testing. Eddy current testing involves the interaction between the external magnetic field of the test coil or probe and the internal magnetic field resulting from the induced currents in the components. When a part is scanned by the probe, defects will cause a change in the impedance of the test coil that is displayed as a signal on the cathode ray screen. However, the evaluation and interpretation of the signals require considerable skill. The equipment must be calibrated using a reference sample of the same material with a known defect and the probe must be specially designed for each application to suit the geometry of the part.

Eddy current testing can be used on any conducting material including non-ferromagnetic materials such as superalloys and stainless steels. The process can be very effective in detecting surface and subsurface cracks in blades and discs and is gaining wider acceptance in the turbine industry. Eddy current can also be used to sort materials and measure coating thickness.

Liquid penetrant inspection (LPI). Liquid penetrant inspection is widely used throughout the turbine industry to detect surface cracks. The process appears to be

deceptively simple and is often misused. Surface preparation is particularly important because defects can easily be masked by overpeening.

In the analysis of service-exposed components such as discs and blades, a judicious compromise is often required on the surface preparation. Light glass bead cleaning is often effective in removing surface deposits with minimal loss in sensitivity; however, care must be taken not to overpeen the surface. Where optimum sensitivity is required, such as in rotating blades, the parts should be chemically etched after glass bead cleaning to remove metal peened over the discontinuities.

LPI is designed to locate only discontinuities that are open to the surface. When a penetrant is applied to the surface of a component, it is drawn into surface discontinuities by capillary forces. After the excess penetrant has been removed from the surface, the penetrant trapped by the defects is drawn back out by capillary action and forms a detectable outline of the defect.

Over the years, many penetrants have been developed that vary in sensitivity and application. As indicated in the enclosed information sheets from the manufacturers, the penetrants are divided into water washable and post-emulsifiable (solvent remover) types. Commonly used penetrants include ZL-17B or Ardrox 970-P10 (water washable) and ZL-22A (solvent type). Care must be taken to ensure that the correct combinations of penetrants, removers, and developers are used. In addition, a test should never be repeated using a different penetrant type because complete masking of the defects will occur.

Field inspections are particularly challenging because the inspection conditions are generally not ideal and often the operator does not have all the equipment desired. Although some compromises are often required, the inspector must ensure that the sensitivity and validity of the test are not jeopardized.

Laser techniques. These can be used for inspection, calibration, and highest-resolution surface mapping.

Computer-aided topography (CAT) scan. CAT scanners locate internal defects in engine components. It is particularly useful for sophisticated components such as blades and airfoils with internal passages and cavities.

Powder metallurgy

This involves molding of powdered alloy flux (somewhat like plasticine) to damaged areas of a component (blades and vanes typically) and use of elevated temperatures to make that flux integral with the parent material. Some sophisticated repair shops develop their own equivalent process and do the work under license to OEMs or sell the OEM (and other facilities) the right to use the process.

Welding

Traditional weld repairs could result in component warpage, which then requires specialized heat treatment and associated jigs, which, in turn, extends repair times and costs, with severe consequences for overall turbine maintenance.

- Laser welding: Extremely accurate and much less heat intensive than conventional repair, laser welding is particularly useful for turbine blades, compressor blade leading edges, and other sensitive components.

- Weld repair using robotics*: The automation of turbine blade welding provides both metallurgical benefits and production advantages. Heat-affected zone cracking in sensitive superalloys, such as IN738 and IN100, can be eliminated or greatly reduced by optimizing process control, and higher production yields can be achieved when welding jet engine blades. However, the successful implementation of automated processes requires careful consideration and engineering of the technology package. In particular, the equipment packager must be experienced in the technology associated with turbine blade welding and incorporate appropriate tooling, measurement system, power source and robotic controls.
- Superalloy welding at elevated temperatures (SWET): OEMs often develop their own proprietary process for this technology that is commonly used on superalloy or directionally solidified materials such as turbine blades.
- Dabber TIG (tungsten inert gas): A slightly older process that uses TIG to rebuild knife edge seals with minimal heat warpage.
- Plasma transfer arc: Similar to dabber TIG and used for the same components.

The exception to the "land-based turbine design approaches aircraft engine technology standards" is evident in certain OEM models, such as Alstom's GT11N2 and GT35. However, that is because both these models are designed to take the punishment meted out by vastly inferior fuels or just be conservative enough to require less training for end-user operators and maintenance staff. Alstom also makes sophisticated models with metallurgy that will match those of the "dual" (both aircraft engine and land-based engine) OEMs for users with different requirements and less punishing fuels. However, Alstom also contracts powder metallurgy repair, for instance.

All OEMs can enhance their "benefits to end user" objectives from some for the preceding techniques, such as the ultimate time-saver laser machining. It ultimately depends on the specifics of an end user's application. The potentially usable repair techniques on any end user's selection play a huge role in determining the turbine model's overall maintenance costs and therefore the ultimate crux of gas turbine selection, called "total costs per fired hour."

Basic Fundamentals of Materials

The properties of the materials used in gas turbines are determined by their composition and their prior processing and service history. To understand how these factors work to govern alloy behavior, a basic understanding of some fundamental principles of materials engineering is useful. This is largely a question of understanding some of the terms used by metallurgists in describing material behavior.

Turbine materials are governed by the laws of thermodynamics, which basically means that changes that take place in the materials result in a reduction of the energy state in the material. We often speak of the *equilibrium* or *stable* condition of a material; this simply means the condition of lowest energy.

Given infinite time, all materials would end up in their equilibrium condition. In

* Source: Adapted from extracts from Lownden, Pilcher, and Liburdi, "Integrated Weld Automation for Gas Turbine Blades," Liburdi Engineering, Canada, ASME paper 91-GT-159.

practice, there are *kinetic* barriers to achieving equilibrium and most materials are used in a *metastable* condition. The most common kinetic barrier is the rate of *diffusion* (i.e., the speed at which atoms in a solid material can rearrange themselves). Almost all of the metallurgical reactions that occur in turbine materials occur at rates governed by the speed of diffusion. Examples include the rate at which a coating interdiffuses with the base metal or the rate at which strengthening particles grow in an alloy.

All of the materials used in gas turbines are *crystalline* in nature. This means that the atoms of the elements that make up the alloy are arranged in regular periodic arrays or lattices, with each atom occupying a site in the array. When we refer to grain or crystal *orientation*, we are referring to the direction relative to this crystal lattice. The mechanical and physical properties of materials depend on their orientation.

In real materials, the crystals are not perfectly periodic, but contain various lattice defects. Two of the most important of these are *dislocations* and *grain boundaries*. Dislocations are an important class of planar defects, since their presence within crystals leads to plastic deformation behavior. Most materials are not used as a single crystal, but as polycrystals that consist of many individual crystals with different orientation called *grains*. The interfaces between the individual grains are *grain boundaries*. The size and degree of orientation between grains and the nature of the grain boundaries are important in determining the properties of a material.

Metallurgists control the nature of grains by the processing performed during manufacture. In a polycrystalline material, grains will increase in size at elevated temperatures and thus *grain growth* will occur during high temperature heat treatments. Grain sizes can be reduced by introducing plastic work into a material. At high temperatures, the resulting strain energy drives the process of *recrystallization*, which results in the formation of smaller grains during heat treatments or hot-working operations.

Engineering materials are almost exclusively mixtures of two or more elements, which are called *alloys*. Alloying elements can dissolve in the matrix of the principal elements to form a *solid solution*, in which the dissolved element is randomly distributed in the crystal lattice. The alloying elements can also react with the matrix to form a *compound* that has a specific arrangement of atoms of each element.

Commonly both solid solutions and compounds will coexist within the same material as different *phases*. The stability of specific phases within a given alloy system varies with the composition and the temperature. Kinetics also determine which phases form within an alloy. Many reactions are sluggish enough that the stable phase may not form initially and the alloy may exist in metastable condition for some length of time. By using chemistry and heat treatment to control the phases formed by an alloy, metallurgists can alter the strength of materials.

Three principal types of deformation take place upon the application of loads to turbine materials. *Elastic* deformation is instantaneous reversible deformation that results from the distortion of the crystal lattice. *Plastic* deformation is the irreversible deformation that takes place instantaneously through the movement of dislocations through the crystal matrix. *Creep* deformation takes place by a variety of diffusion-controlled processes over time, resulting in continuing strain under the applied load.

At sufficiently high loads or after a critical amount of deformation has taken place, *fracture* of a material will occur. Fracture can be broadly classed as ductile or brittle. In turbine materials under most conditions, fracture occurs by the

formation and linkage of internal cavities formed either by creep or plastic deformation.

When cyclic loads are applied to a material, cracking and fracture may occur by the process of *fatigue*. On a microscopic scale, fatigue occurs by localized plastic deformation, resulting in the initiation and growth of macroscopically brittle cracks.

The resistance of a material to deformation and fracture depends on its composition and *microstructure*. The size of grains, the size and distribution of second phases, and the effects of alloying elements on the crystal lattice of the matrix all influence the mechanical behavior of the alloy.

Exposure of alloy surfaces to operating conditions results in surface reactions between the alloy and the environment. *Oxidation* and *hot corrosion* occur at elevated temperatures through direct reaction with oxygen and other environmental contaminants. *Aqueous corrosion* occurs in wet environments through dissolution reactions.

Material Selection for Gas Turbines

This subsection is with specific reference to gas turbine materials, the most severe thermal application in a plant.

The materials used in gas turbines or jet engines span the range of metallurgical alloys from high-strength steel, to lightweight aluminum or titanium, to temperature-resistant nickel or cobalt superalloys. In a gas turbine, the temperatures can vary from ambient to gas temperatures in excess of the melting point of superalloys and, therefore, the materials in the different sections must be selected on the basis of their capability to withstand the corresponding levels of stress and temperature. The following summary outlines the materials used in the different components of the gas turbine, along with a rationale for their selection.

Compressor rotor

The temperature in a typical compressor will range from ambient to approximately 800°F (425°C). The discs and blades rotate at high-speed and are, therefore, highly stressed and subjected to aerodynamic buffeting or fatigue. In industrial turbines, the discs are generally made from high strength alloy steel and the blades from martensitic stainless steel. However, in jet engine derivatives, lighter materials such as aluminum and titanium are used for the blades and vanes in the front of the compressor. In some cases, the last stages of the compressor can run significantly hotter and more creep-resistant materials must be used such as A286 and IN718.

Turbine discs

Turbine discs are highly stressed in the rim area where the blade root attachment occurs and in the hub of bored discs where high burst strength is required. The discs are forged from high-strength steels in advanced industrial turbines and iron or nickel base superalloys such as A286 and Inconel 718 for the jet engines. The disc rim is generally isolated from the hot gas path and cooled to as low as 600°F (315°C) for alloy steel discs to ensure adequate material strength and creep resistance.

Combustion cans

The flame temperature in a burner generally exceeds 3000°F (1650°C). The temperature is moderated by mixing with cooler compressor discharge air that flows

around the combustion chamber and through the slots in the walls to keep metal relatively cool [approximately 1500°F (815°C)]. The combustor cans are generally fabricated from nickel base sheet superalloys such as Hastelloy X, Nimonic C263, and Inconel 617. These alloys have good weldability and oxidation resistance.

Turbine vanes

The stationary vanes in a turbine act as guides for the hot gas to ensure that it enters the blade's airfoil at the right angle and with minimal pressure loss. The applied stresses are generally low; however, they are subjected to turbine inlet gas temperatures which, in some engines, exceed the melting point of the material [2500°F (1370°C)]. The vanes are generally made from cobalt base alloys such as FSX414 and X45, which have good castability and excellent oxidation and thermal shock resistance. In advanced designs, the vanes are cast with integral cooling passages to reduce the metal temperature. The cobalt base alloys are generally weldable and minor weld repairs are often allowed. In some designs, the cobalt base alloys have been replaced with more creep-resistant nickel base alloys such as IN738, Rene 80, and IN939, which are significantly harder to weld repair.

Turbine blades

Turbine blade airfoils are subjected to the most severe combination of applied stresses due to centrifugal and bending loads and high temperatures. The blade materials must have excellent strength and creep resistance, as well as oxidation resistance. In advanced units, the blades are cooled by internal passages to moderate the metal temperature and improve blade life. The blades are generally precision forged or cast from nickel base alloys such as Udimet 520, IN738, Rene 80, and Mar M247 and can be manufactured as either equiaxed, directionally solidified, or single crystal castings. These materials have poor weldability and repairs must be approached with extreme caution.

Service life for turbine components

Once in service, critical gas path components require care and attention to optimize their life potential. Often, turbine users choose to abdicate their responsibility in this critical area and elect to rely solely on the manufacturer for guidance. However, this can lead to premature component replacement of failures if components are neglected or improperly assessed.

A gas path component management program basically involves the detailed characterization of components at established intervals. For example, during major overhauls, representative blades are removed and destructively tested for corrosion, microstructure, and remaining creep life. The data are tabulated and a life trend curve established for the material. This will provide the user with specific information on how the engines are standing up rather than rely on someone else's data and provide advance warning of impending problems with corrosion or creep.

Creep damage can be detected in turbine blades by judicious testing of samples from the airfoil and by metallurgical inspection. With prolonged service exposure at high temperature and stress, cavities are formed at the grain boundaries of the material that, with time, will grow in number and size and eventually join to form a crack. However, if creep voiding can be detected prior to surface crack formation, the parts can be rejuvenated by hot isostatic pressing (HIP).

During HIP, the parts are subjected to a combination of pressure and temperature that collapses any internal cavities and diffusion bonds the surfaces back together. Tests have shown that, in most materials, HIP is capable of fully restoring the properties and, therefore, offers the opportunity to recycle and extend the life of serviced blades.

Steam-turbine metallurgy

Steam turbines operate in far less demanding service than gas turbines as the following list of typical steam turbine materials illustrates.

Typical steam turbine materials

Part Identification	Material	Commerical Equivalent
Steam rings — 650 psig/750°F	Carbon steel	ASTM A216 grade WCB
— 900 psig/950°F	Alloy steel	ASTM A217 grade WC6
— 1500 psig/950°F	Alloy steel	ASTM A217 grade WC9
Valve chests — 650 psig/750°F	Carbon steel	ASTM A216 grade WCB
— 900 psig/950°F	Alloy steel	ASTM A217 grade WC6
— 1500 psig/950°F	Alloy steel	ASTM A217 grade WC9
Cylinders	Carbon steel	ASTM A216 grade WCB
Nozzle blocks	Carbon steel	ASTM A212 grade A
	Stainless steel	AISI 416
Sovernor valve	Stainless steel	AISI 410
Sovernor valve stem	Stainless steel	AISI 410
Shafts	Alloy steel	AISI E4340
Discs — standard forged	Alloy steel	ASTM A294 class 5
— integral with shaft	Alloy steel	ASIS E4340
Blading	Stainless steel	AISI 403
Bearing brackets	Carbon steel	ASTM A216 grade WCB

Manufacture

Superalloys

The term superalloy is popularly used to define a material having structural strength above 1000°F (540°C). The development of these temperature-resistant materials in the early 1940s was a primary catalyst for jet engine development. The two technologies have been interrelated ever since and, as the temperature capability of materials was improved, so did the turbine efficiency. The enclosed graph illustrates the strength improvements obtained through alloy development and how the chemistry and microstructure became more complex with each evolution.

Basically, superalloys can be categorized into three chemical families: iron base, nickel base, and cobalt base. All alloys contain additions of chromium and aluminum for oxidation resistance and variations of aluminum, titanium, molybdenum, and tungsten for strength. The nickel base superalloys offer the highest strength and are generally chosen for rotating blade applications. The cobalt base alloys, although weaker, offer better environmental resistance and are generally chosen for stationary vane applications.

From a metallurgical point of view, all superalloys exhibit an austenitic or γ microstructure that, unlike steel, does not undergo any transformation as it is heated to its melting point. The alloys derive their strength from three principal mechanisms:

1. Solid solution hardening of the γ matrix resulting from the coherency strains or stiffening effect or larger atom elements such as chromium, molybdenum, and tungsten.
2. Precipitation of small γ' particles Ni_3 (Al, Ti) throughout the matrix act as obstacles to dislocations or strain flow. The γ' phases precipitate on cooling from the solution temperature and during aging treatments.
3. The formation of blocky carbides at the grain boundaries act to pin the grain boundary and prevent grain boundary sliding during creep.

Manufacture: Castings and Forgings

There are two main routes by which superalloys are manufactured into turbine components: casting and forging.

Forged parts are made by hot-working cast ingots or powder metallurgy compacts into billet or bar and eventually into the final shape. Since it involves significant plastic deformation, considerable refinement of grain size can be achieved through recrystallization; however, the alloy must also have good ductility at the forging temperature. Consequently the higher strength superalloys cannot be manufactured by forging. Turbine discs, some turbine blades, and compressor components are made by forging.

Cast parts are formed by pouring molten alloy into near-net-shape molds and allowing them to solidify. The investment casting process used allows extremely complex shapes, including cooling passages, to be cast in. Molds are made by depositing a ceramic layer around a wax form and melting the wax to form a cavity, while ceramic cores are used to cast internal passages. Because the process does not involve the deformation of the alloy, ductility is not an important consideration. Turbine blades and vanes are commonly made by casting.

A modification of the investment casting process is used to produce directionally solidified and single crystal components. To produce components in these forms, alloy is cast into molds that are subsequently withdrawn from a furnace at a controlled rate. By controlling the solidification of the casting, the grains are forced to grow in one direction and by using a crystal selector at the base of the casting, the casting can be made as a single crystal. Such components have improved creep and thermal fatigue resistance because there are no grain boundaries oriented perpendicular to the principal stress direction. Because of their high cost, such parts are typically limited to first-stage turbine blading.

Metering, Fluids; Metering Pumps (*see* Fuel Systems)

Mist Eliminators (*see* Separators)

Mixers (*see* Agitators; Centrifuges)

Monitoring (*see* Condition Monitoring)

Motors (*see* Electric Motors)